Mutational Signatures

Massimo Maiolo

23.10.2025

Institute of Tissue Medicine and Pathology – University of Bern

Github Cancer Genomics Course

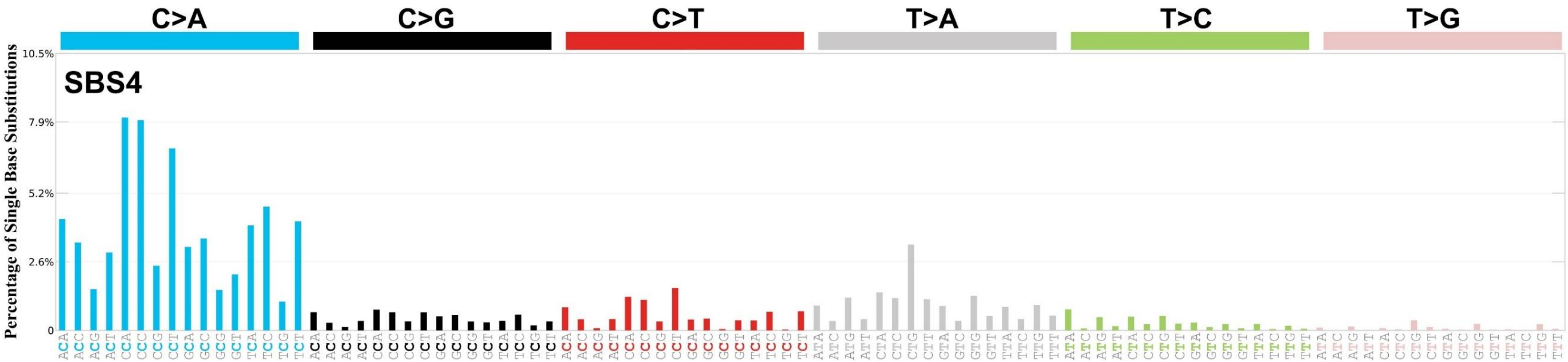
https://github.com/supermax1234/CancerGenomicsCourse

Understanding Mutational Signatures

What is a Mutational Signature?

- a **specific pattern of mutation** types in DNA
- **caused** by a **particular biological process**, like a **mutagen** or a **cellular defect**

SBS 4 : tobacco smoking signature



Proposed aetiology

Associated with **tobacco smoking**. Its profile is similar to the mutational spectrum observed in experimental systems exposed to tobacco carcinogens such as benzo[a]pyrene. SBS4 is, therefore, likely due to direct DNA damage by tobacco smoke mutagens.

https://cancer.sanger.ac.uk/cosmic

Damage, Repair, and Replication

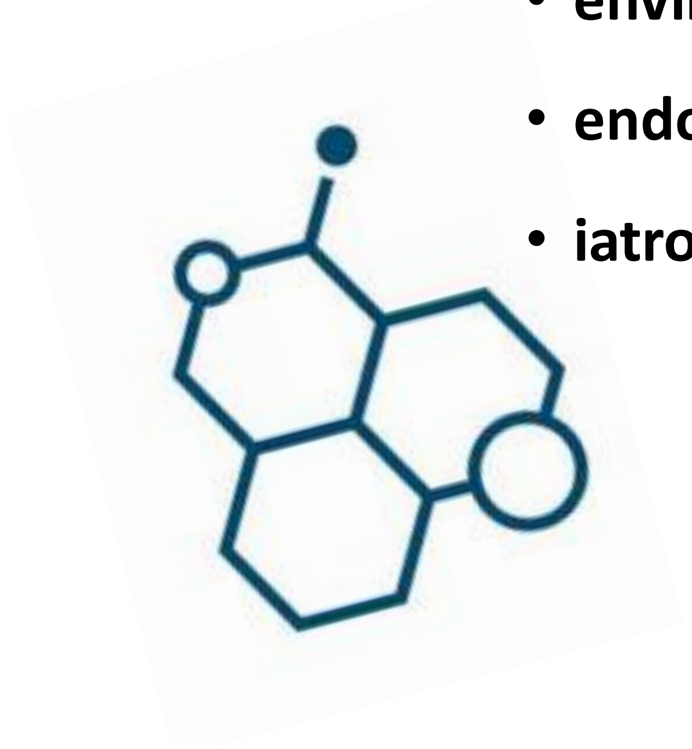
Mutational signatures are **categorized** based on the **underlying processes** that cause the DNA damage, how the cell tries to repair it, and errors made during replication.

Understanding these three main categories — **Damage, Repair, and Replication** — is crucial because it allows us to **link** a mutation pattern directly to a specific **biological or environmental cause**.

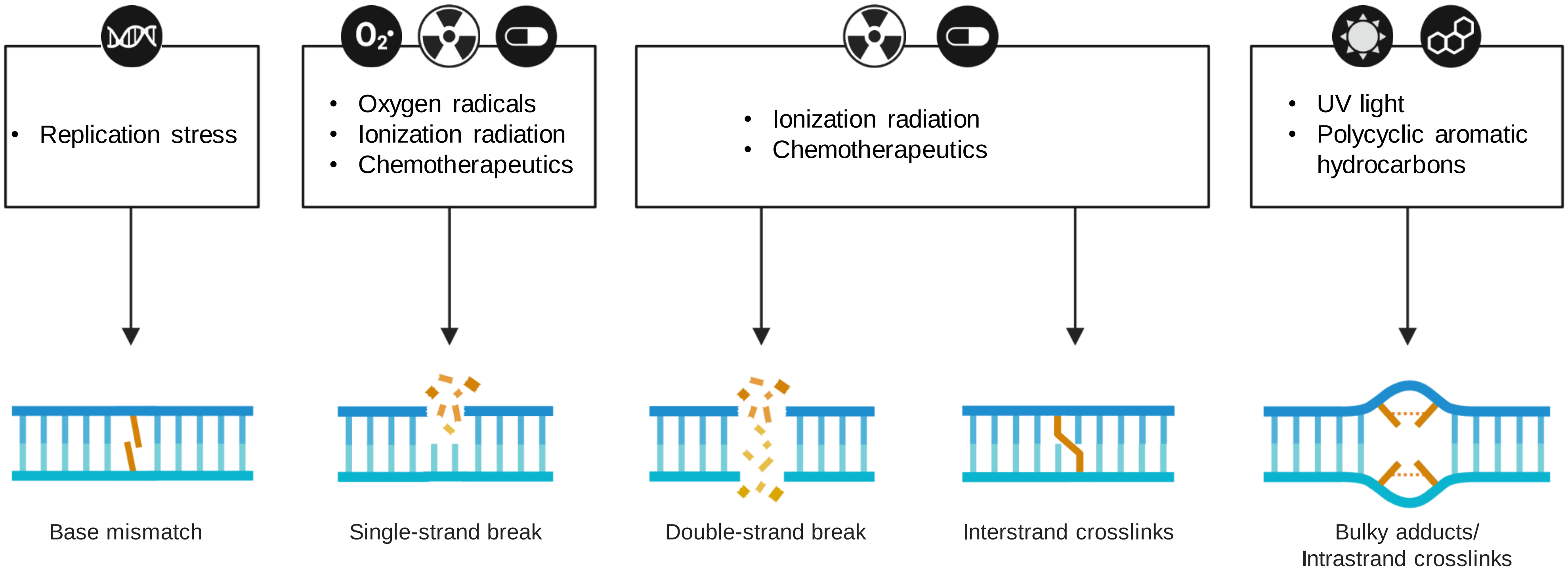
Mutagenic factors

Mutagenic factors are agents that can cause mutations by damaging DNA. These factors can be broadly classified into three categories:

- **environmental**
- **endogenous** (internal to the body)
- **iatrogenic** (resulting from medical treatment)



Common Causes of DNA Damage



Environmental Mutagens

- **Tobacco Smoke**

  Contains a complex mixture of chemicals, including polycyclic aromatic hydrocarbons (PAHs), which are potent mutagens that can cause specific mutational signatures in lung, bladder, and other cancers

- **Ultraviolet (UV) Light**

  The UV radiation from sunlight or tanning beds is a primary cause of melanoma and other skin cancers. It creates characteristic "UV damage" signatures on the DNA

- **Aflatoxin**

  A toxic compound produced by certain molds found on food products like corn, peanuts, and tree nuts. It is a major cause of liver cancer, particularly in regions with poor food storage practices

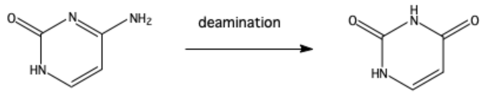
- **Aristolochic Acid**

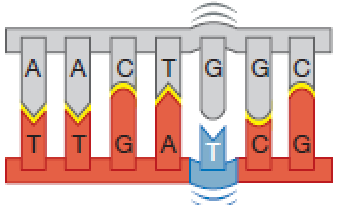
  Found in plants of the Aristolochia genus, often used in traditional herbal medicines. It is a powerful carcinogen linked to kidney and liver cancers

- **Air Pollution**

  Particulate matter and other pollutants in the air contain various carcinogens that can contribute to lung cancer

Endogenous Mutagens





- **Reactive Oxygen Species (ROS)**

  Produced during normal metabolic processes, ROS can damage DNA, leading to mutations

- **Spontaneous Deamination**

  This is a natural chemical reaction where DNA bases can be altered, such as cytosine changing to uracil, which can lead to mutations during DNA replication

- **DNA Replication Errors**

  While DNA polymerase has proofreading capabilities, mistakes can still occur during the replication process, leading to a natural background rate of mutations

Iatrogenic Mutagens

- **Chemotherapy**

  Many chemotherapy drugs, such as platinum compounds, are designed to damage DNA and kill cancer cells. As a side effect, they can also induce new mutational signatures in the surviving cells, sometimes leading to therapy-related cancers.

- **Radiation Therapy**

  Similar to UV light, high-energy radiation used in cancer treatment can cause extensive DNA damage and introduce unique mutational patterns in the exposed tissue.

DNA Repair Signatures (Deficiency)

These signatures arise not from a direct external damaging agent, but from the **failure** or **absence** of a **critical DNA repair pathway**. When the primary, high-fidelity repair mechanism fails, the cell is forced to use an error-prone *backup* pathway, generating a characteristic pattern.

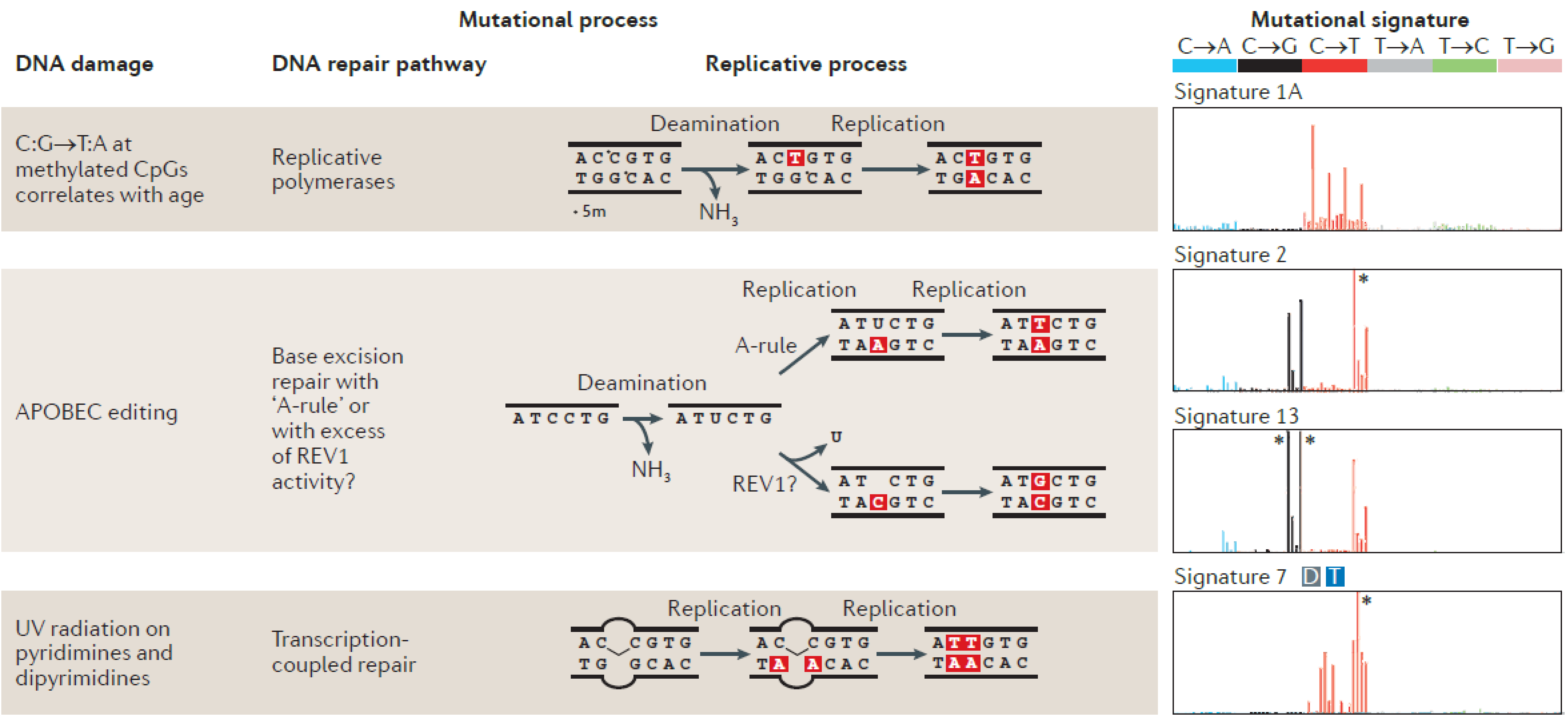
- The **lack of repair fidelity** generates the signature, often affecting the entire genome rather than localized sites.
- **Examples**:
  - **Homologous Recombination Deficiency** (**HRD**) (SBS3): Associated with mutations in genes like **BRCA1/2**. When HR fails, the cell uses error-prone pathways like non-homologous end joining (NHEJ), resulting in large genomic deletions and specific mutational patterns.
  - **Mismatch Repair Deficiency** (**MMRd**) (SBS15/20/26): Failure of the MMR system (often due to MSH2/6 or MLH1/PMS2 defects) leads to high mutation burdens, particularly at repetitive sequences (microsatellite instability).

DNA Replication Signatures (Polymerase Errors)

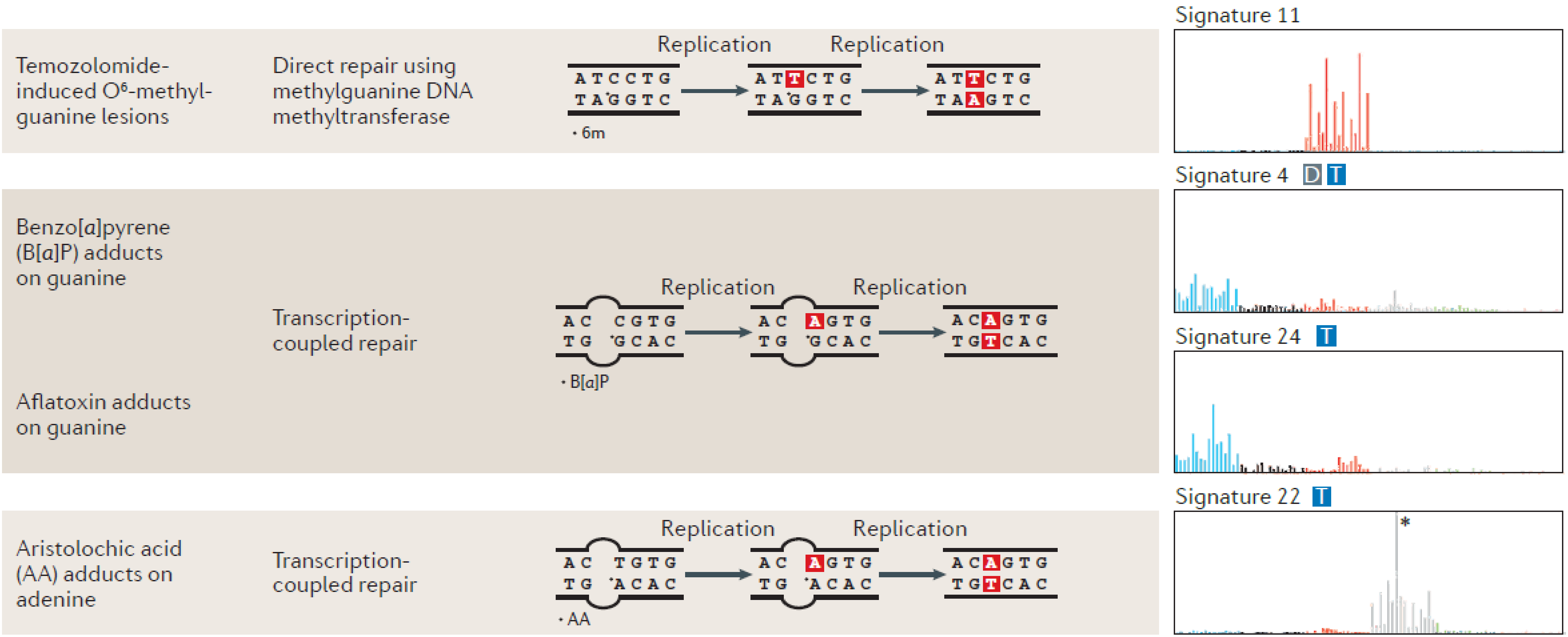
This category involves processes that occur during the **DNA synthesis (replication)** phase. The signatures are caused by **errors** in the DNA **polymerases** (the enzymes that copy the DNA) or the proofreading mechanisms.

- These are **errors in the DNA copying machinery** itself, independent of external damage.
- **Examples**:
  - **Polymerase ε Deficiency (SBS10a/b):** Mutations in POLE lead to a profound lack of proofreading during replication. This results in the highest known mutation rates (hypermutation) with highly specific **C>A** and **C>T** patterns, often seen in endometrial and colorectal cancers.
  - **Polymerase δ Deficiency (SBS14):** Less frequent than Pol ε, but also results from defective polymerase proofreading.

Examples of mutational processes



Helleday et al, Nat Rev Genet 2014



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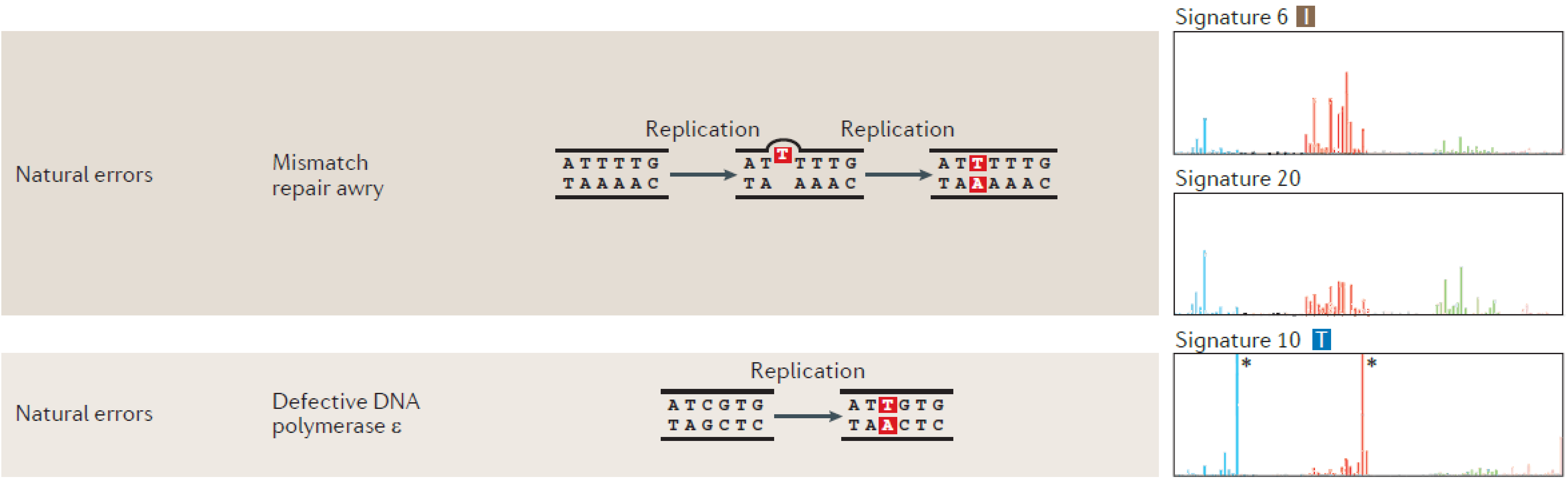


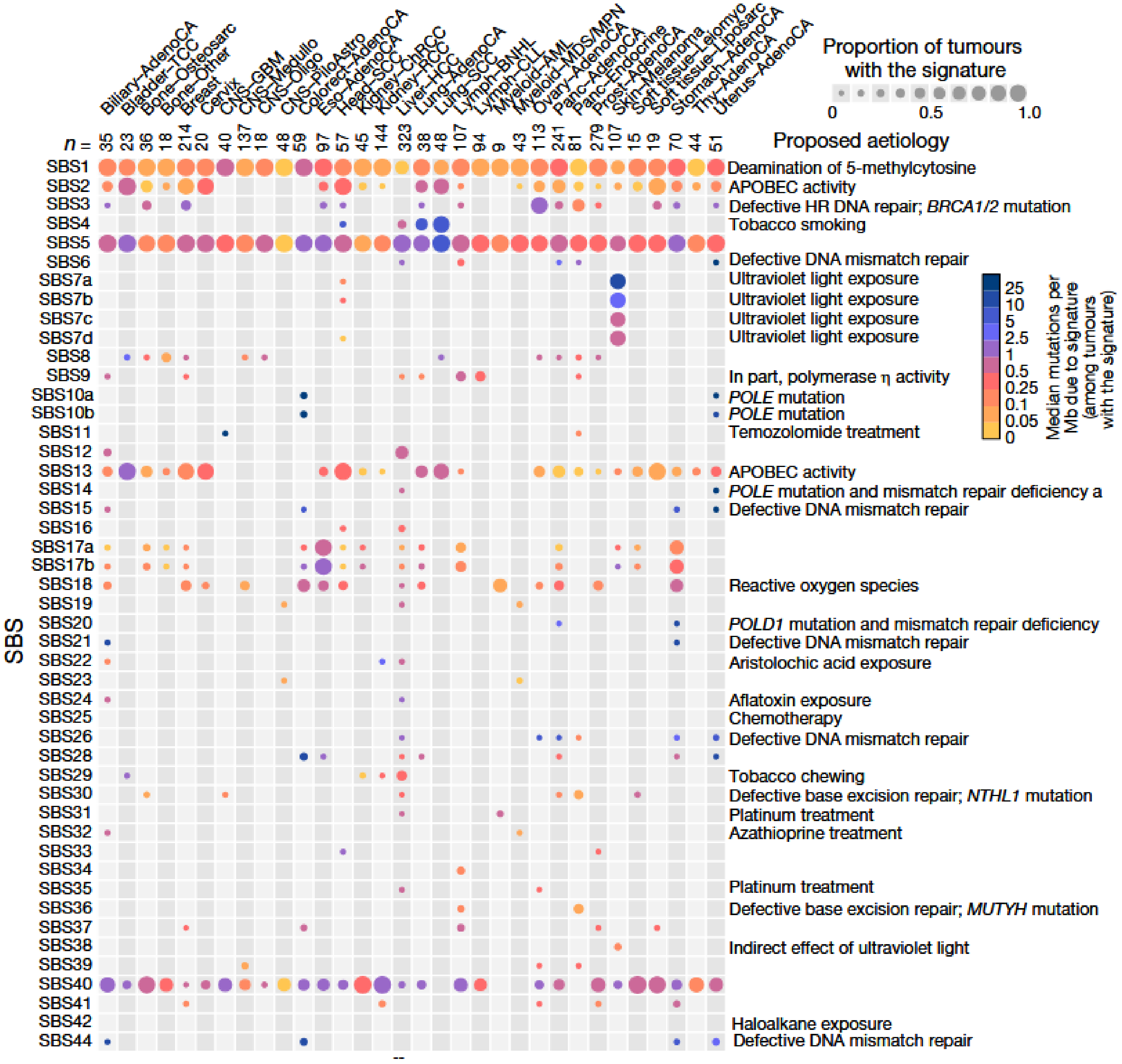
Figure 2 | **Summary of known mutational signatures, and the components of DNA damage and repair that constitute the mutational processes.** There are marked differences among the 96-element mutational signatures, which are dominated by specific elements, including enrichment of various base substitutions (shown in the graphs on the right), transcriptional strand bias (T), excess of dinucleotide mutations (D), and association with insertions and deletions (I). The asterisks mark instances at which the limits of the *y* axes, which represent the likelihood of specific mutations being present in a signature, are exceeded. 5m, 5′ methyl group; 6m, O^6 methyl group; APOBEC, apolipoprotein B mRNA editing enzyme, catalytic polypeptide; REV1, DNA repair protein REV1; UV, ultraviolet.

Helleday et al, Nat Rev Genet 2014

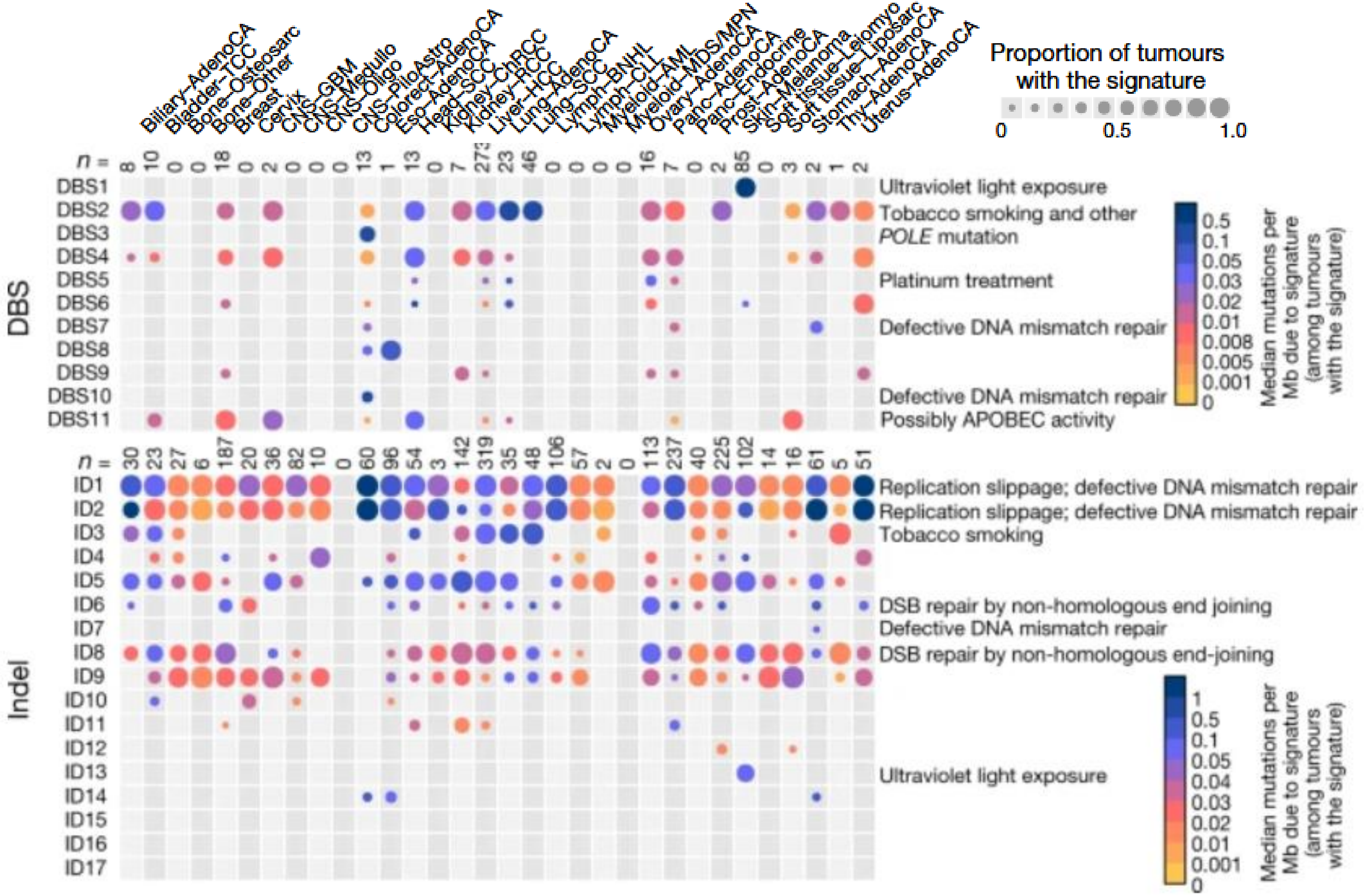
Signatures Need Context

Not a 1-to-1 Match

- A signature is usually **not unique** to a **single tumor type** or a **specific mutagen**
- Many **different causes** can lead to the **same signature**



https://www.nature.com/articles/s41586-020-1943-3 -



https://www.nature.com/articles/s41586-020-1943-3 -

Why mutational signatures matter

Mutational signatures matter because they act as a forensic record of a cell's history, **revealing the processes that have damaged its DNA**.

By analyzing these fingerprints left on the genome, scientists and doctors can gain crucial insights into the causes of a disease, particularly cancer.

Why mutational signatures matter

- **Uncovering the Causes of Cancer**

  Mutational signatures can help scientists understand what causes cancer. Each signature is linked to a specific process. For example, a signature caused by ultraviolet (UV) radiation is found in melanoma, while another is linked to cigarette smoke and is found in lung cancer. This provides direct evidence of the role of these factors in tumor development.

- **Improving Cancer Diagnosis**

  Signatures can provide more information than a standard diagnosis. For instance, two tumors might look identical under a microscope, but their mutational signatures could be very different. This difference can reveal a lot about the tumor's origin, which can help in choosing the right treatment.

Why mutational signatures matter

- **Guiding Treatment Decisions**

  Mutational signatures can help predict how a patient will respond to certain therapies. Some signatures indicate that a tumor will be sensitive to specific drugs. For example, a signature caused by a defect in DNA repair (e.g., in BRCA-mutated breast cancer) suggests that the tumor will likely respond well to targeted therapies like PARP inhibitors.

- **Monitoring Treatment Response**

  In some cases, changes in mutational signatures over time can be used to track a tumor's evolution and its response to treatment. If a new signature appears, it might indicate that the tumor is developing resistance to a drug, allowing doctors to adjust the treatment plan.

- Mutational signatures **transform** our understanding of **genetic damage** from a simple list of changes into a rich, informative story about **how** and **why** a **disease developed**.

- They are a powerful tool for **precision medicine**, helping to make diagnosis more accurate and treatment more effective.

From Accumulation to History: The Power of the Pattern

- All cancers **accumulate somatic mutations** -> the pattern of mutations is informative.
- *Mutational signature* : the **characteristic imprint** left by a mutational process (DNA damage + repair/replication) on the cancer genome.
- By identifying signatures we can infer **past exposures**, **ongoing processes**, and **possibly therapeutic vulnerabilities**.

The signature transforms a list of mutations into a story of damage, survival, and potential treatment

Active mutational processes

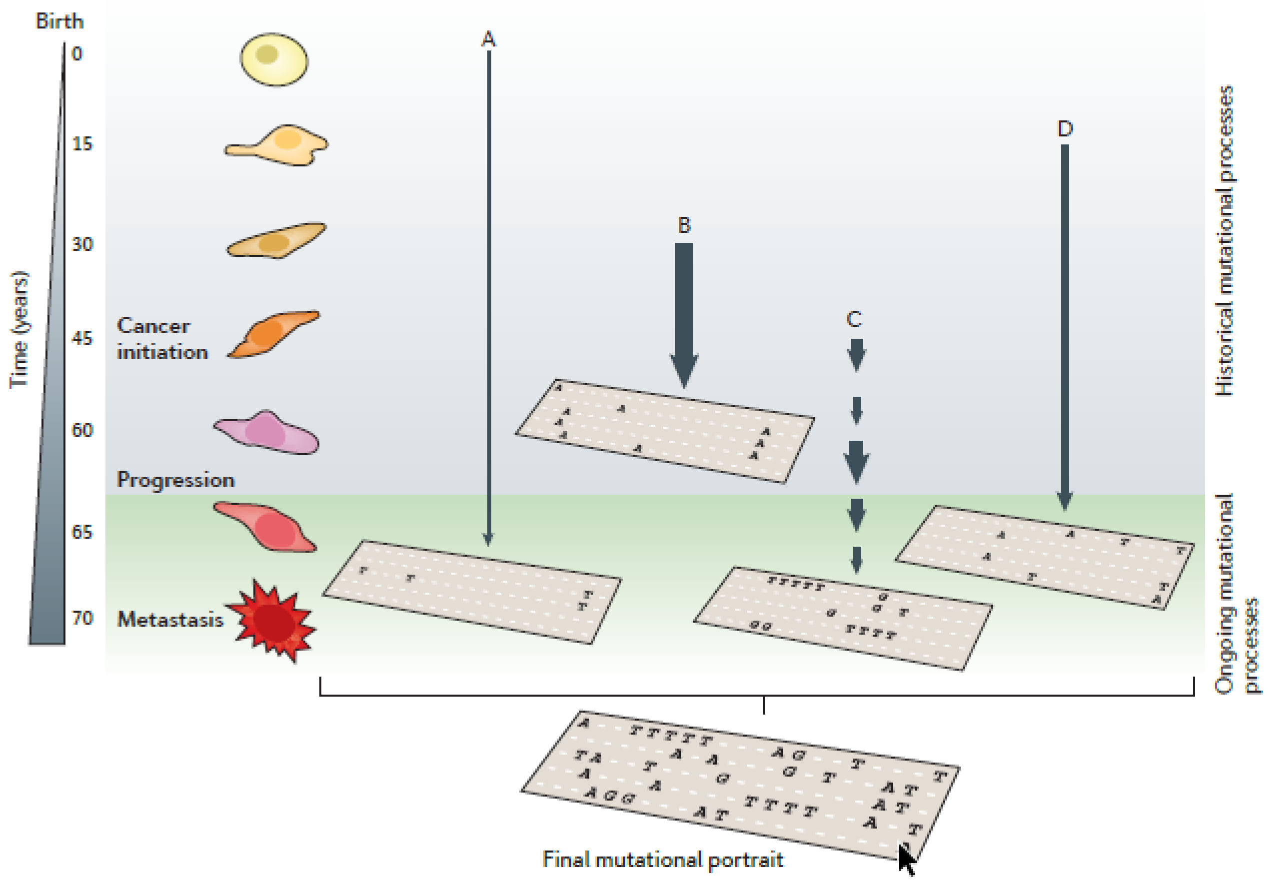


Figure 1 | **Active mutational processes over the course of cancer development.** Each mutational process leaves a characteristic imprint — a mutational signature — in the cancer genome and comprises both a DNA damage component and a DNA repair component. In this hypothetical cancer genome, arrows indicate the duration and intensity of exposure to a mutational process. The final mutational portrait is the sum of all of the different mutational processes (A–D) that have been active in the entire lifetime. Ongoing mutational processes reflect active biological processes in the cancer that could be exploited either as biomarkers to monitor treatment response or as therapeutic anticancer targets. By contrast, historical mutational processes are no longer active. Signature A represents deamination of methylated cytosines, which is ongoing through life. Signature B can be matched up with the signatures of tobacco smoking. Signature C can represent bursts of APOBEC (apolipoprotein B mRNA editing enzyme, catalytic polypeptide)-induced deamination, and Signature D represents a DNA repair pathway that is awry.

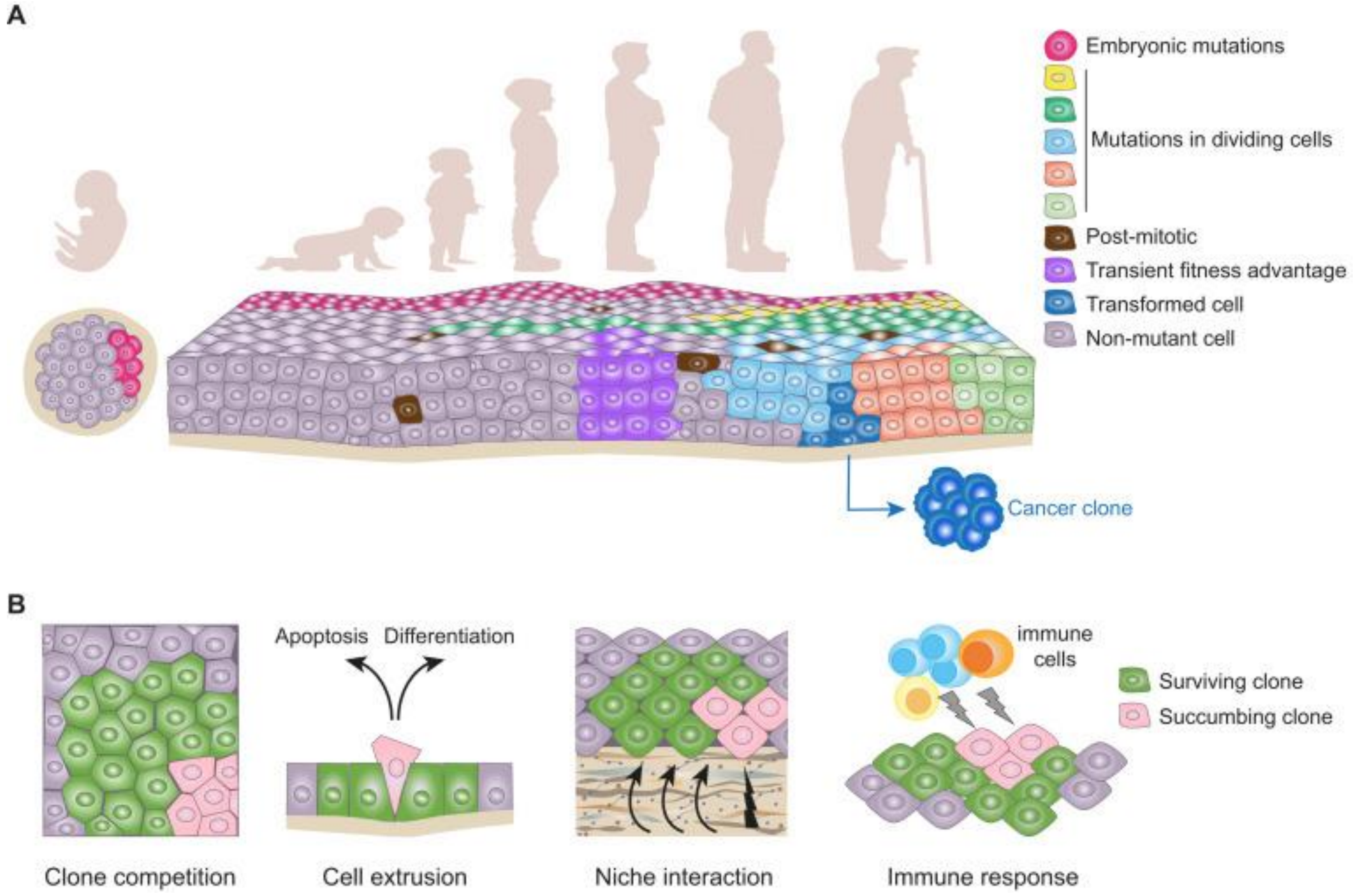
Mutational signature
The pattern of mutations produced by a mutational process.

Mutational portrait
The total genetic changes observed in a cancer genome; that is, the sum of all mutational signatures occurring in a lifetime.

Base substitutions
A type of mutation in which one base is replaced by another in DNA.

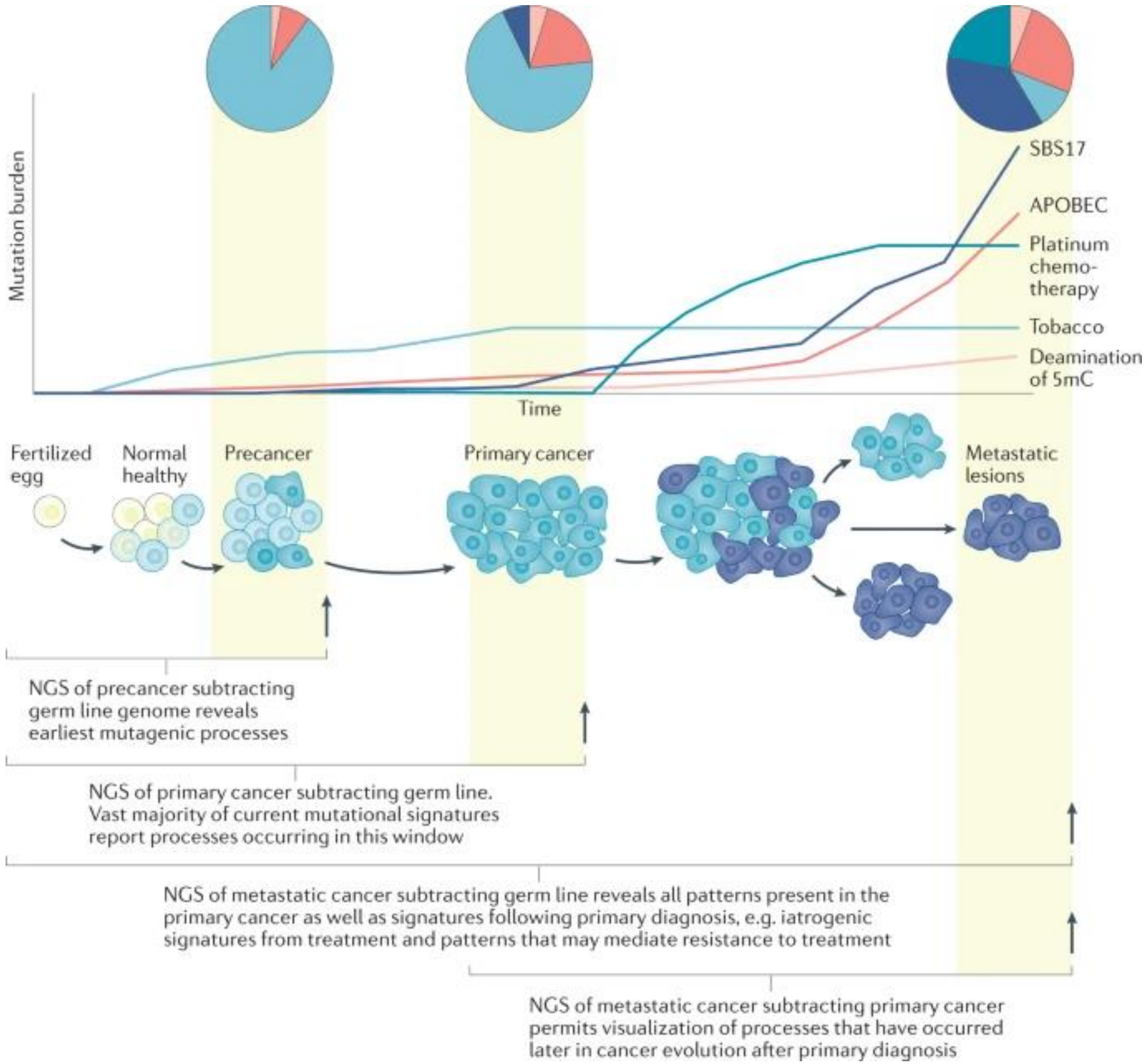
Insertions and deletions
(Indels). A type of mutation that arises from the insertion or deletion of one or more nucleotides within a DNA sequence.

Helleday et al, Nat Rev Genet 2014



Origin, evolution, and fates of somatic clones. (A) Somatic mutations accumulate throughout life resulting in somatic mosaicism. Some mutations are acquired during embryo development and fixed by neutral drift. Other mutations arise in post-mitotic or actively dividing cells due to the exposure to endogenous or exogenous insults. Mutations that confer fitness advantages initiate clonal expansion of actively dividing cells. In some cases, these advantages are transient and the clone disappears when the insult is removed. Some mutant cells may acquire transforming potential and start the tumorigenic process. (B) Clone selection and expansion are driven by several intrinsic and extrinsic factors such as competition between mutant cells with different fitness, cell extrusion leading to apoptosis or differentiation, active responses of the stromal niche, and selective pressure of the immune system.

https://doi.org/10.1016/j.annonc.2022.09.156

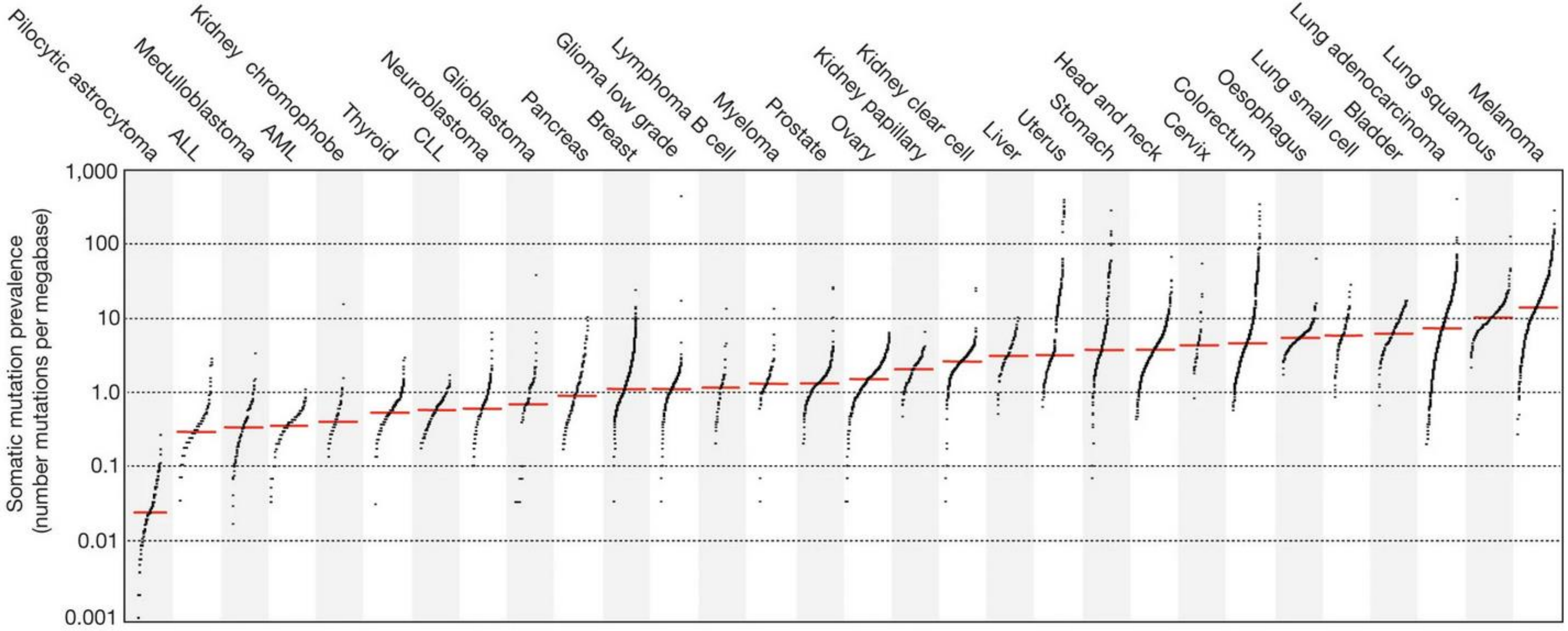


Schematic representation of how a polyclonal population of cells can evolve into a dominant precancer clone, and then a frank carcinoma before metastasis. The graph shows how the mutation burdens of five different mutational signatures change over time. Single-base substitution mutational signature 1 (SBS1) (pink line; caused by deamination of 5-methylcytosine (5mC)) is accumulating in a linear way in the early stages. Tobacco exposure (pale blue) from early adulthood leaves a marked mutational pattern with high mutation counts. However, there is cessation of accumulation of this mutational signature when the patient stops smoking in mid-adulthood. Before formation of the precancerous neoplasm, there is an increase in apolipoprotein B mRNA-editing enzyme, catalytic polypeptide-like (APOBEC)-related mutagenesis (red). Sequencing of the precancerous lesion and subtraction of the germ line would reveal the earliest mutational processes in tumorigenesis (leftmost pie chart). At the point of primary cancer diagnosis, there is a modest increase in APOBEC signatures and the presence of SBS17 (blue line, middle pie chart). When the patient presents with metastatic disease, the sequencing of this disseminated lesion and subtraction of the germ line will reveal all the processes that have contributed to the metastases, including all the processes present in the primary cancer. By contrast, subtraction of the primary cancer could reveal insights that are specific to the post-primary window, including iatrogenic signatures from treatment (for example, dark turquoise line; caused by exposure to platinum chemotherapy), immune-genomic patterns and signatures that mediate treatment resistance. NGS, next-generation sequencing.

https://www.nature.com/articles/s41568-021-00377-7

Clinical applications : examples

- **Cancer etiology**
    - Link cancer type to environmental exposure (e.g. SBS4 in lung → smoking history).
    - **Determine DNA Repair Status:** Use signatures to confirm a functional deficiency in key repair pathways (e.g., **SBS3** is a direct biomarker for **HRD**), which is critical for guiding targeted therapy.
- **Diagnosis & classification**
    - Signatures help define molecular subtypes (e.g. MSI-high colorectal cancers have SBS6/15).
    - Can confirm tumor type when histology is ambiguous.
- **Prognosis**
    - Certain signatures correlate with better/worse outcomes.
    - Example: HR-deficiency (SBS3) predicts sensitivity to platinum therapy.
- **Therapy guidance (Precision Oncology)**
    - BRCA/HRD signatures (SBS3): PARP inhibitors (Olaparib, Niraparib).
    - MMR-deficiency (SBS6/15): immune checkpoint inhibitors (anti-PD1).
    - Chemotherapy exposure (SBS11, SBS31): signatures of past treatment can be identified.



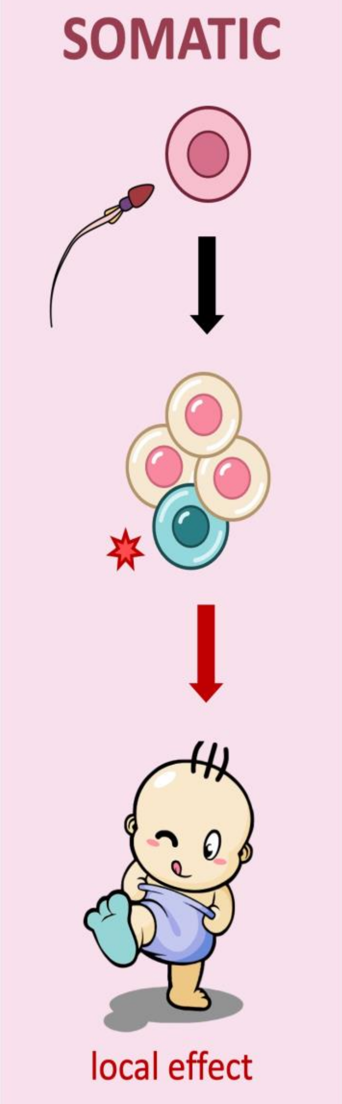
Every dot represents a sample whereas the red horizontal lines are the median numbers of mutations in the respective cancer types. The vertical axis (log scaled) shows the number of mutations per megabase whereas the different cancer types are ordered on the horizontal axis based on their median numbers of somatic mutations. We thank G. Getz and colleagues for the design of this figure[26]. ALL, acute lymphoblastic leukaemia; AML, acute myeloid leukaemia; CLL, chronic lymphocytic leukaemia.

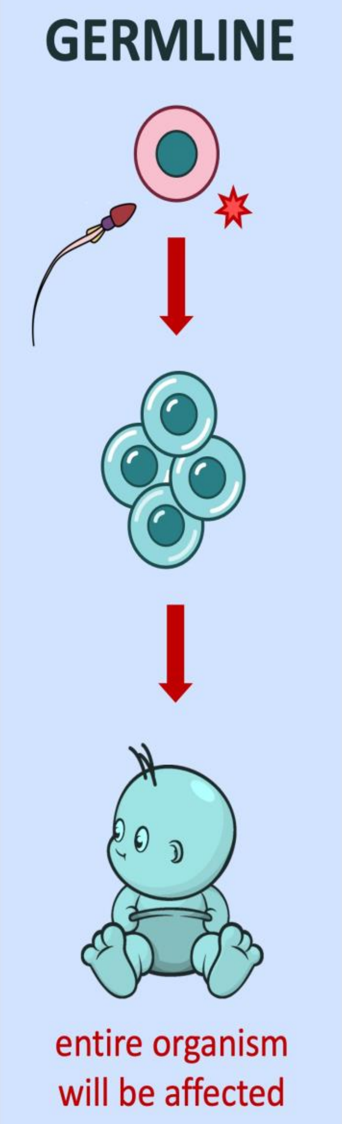
From: Signatures of mutational processes in human cancer

Concepts and Definitions

- **Somatic** mutations vs **germline** mutations
- Types of mutations: **SNVs**, **indels**, **structural variants**
- The **96 trinucleotide context**
- **Mutational catalog**: representation of counts/frequencies

Somatic mutations vs germline mutations





Somatic mutation	Germline mutation
It is defined as any alteration in the genetic sequence of genes of the somatic cells.	It is also called "acquired mutation" as it is acquired during an individual's life.
It is defined as any alteration in the genetic sequence of genes of the germinal cells.	It is also called "hereditary mutation" as it is passed to offspring.
It occurs in somatic or body cells.	It occurs in germ cells.
It is not inheritable.	It is inheritable.
It affects only mutated cells and their progeny.	It affects all the cells of the organism.
It can occur at any stage of the life cycle.	It occurs only during gametogenesis.
It plays no role in evolution.	It is the basis of evolution.
It does not cause genetic disorders but may cause cancer.	It is responsible for genetic disorders and also germline cancers.
It is finished along with the death of the individual.	It continues till the offspring of the individual breed and may result in separate sub-species. Hence it is genetically more important.
In most cases, they show observable effects.	In most cases, they are silent and don't show detectable effects.
They can be treated or cured.	They can't be treated or cured.
Mosaicism occurs.	Usually, mosaicism doesn't occur.
Examples include cancers, tumor development, a neurodegenerative disorder, etc.	Examples include Hemophilia, Down's Syndrome, 18-Trisomy, etc.

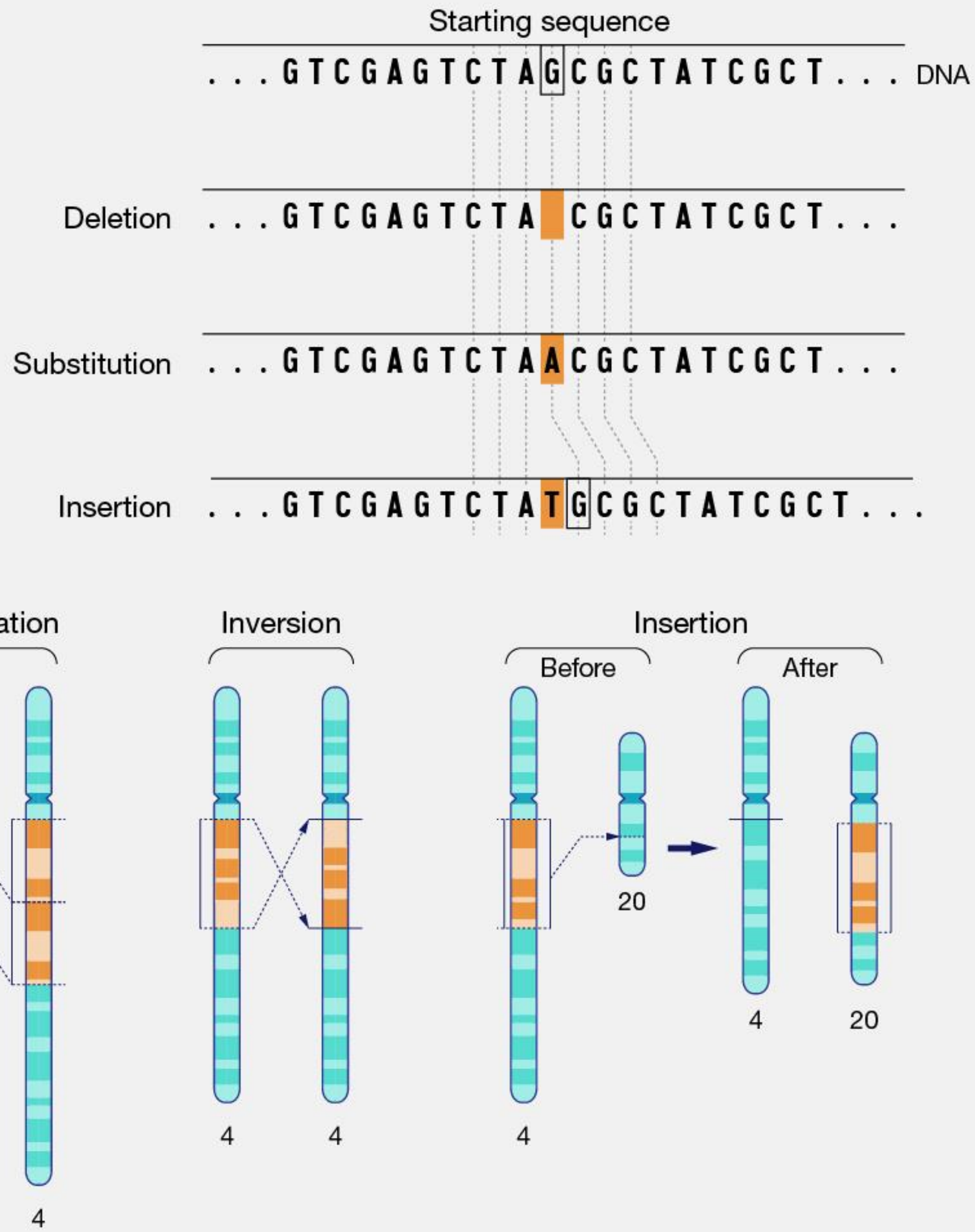
Somatic mutations vs germline mutations

Germline mutation filtering

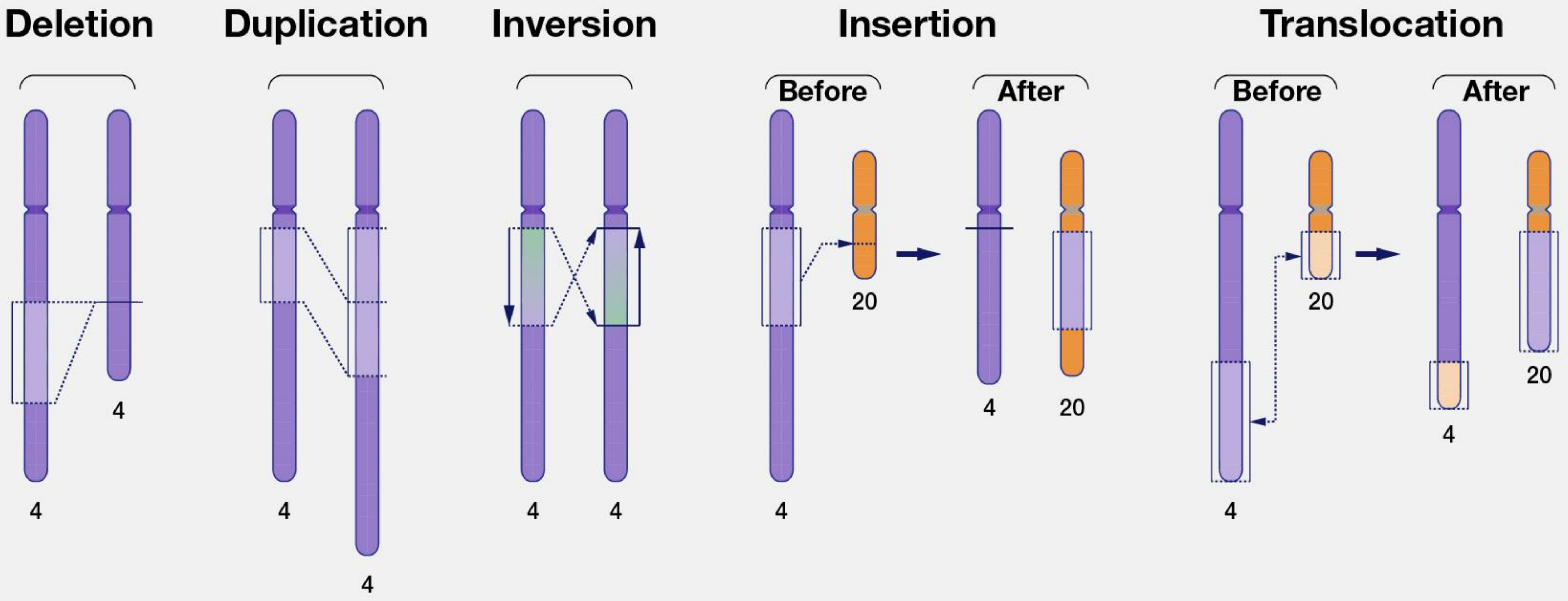
- For mutational signature analysis, the fundamental principle is to **filter out all germline variants** and focus exclusively on somatic mutations.
- **Germline Mutations**
  - *These are genetic changes present in every cell of an organism. They are inherited from parents and are not associated with a specific disease process, such as cancer.*
- **Somatic Mutations**
  - *These are genetic changes that occur after conception in a single cell and are subsequently passed on to its daughter cells. They are not inherited and are typically caused by environmental factors, endogenous processes (like DNA replication errors), or specific mutational processes that lead to diseases like cancer.*
- **If you were to include germline variants in your analysis, you would be mixing inherited, non-cancer-related mutations with the acquired, disease-causing somatic mutations. This would completely obscure the subtle patterns that define mutational signatures, making it impossible to accurately identify and attribute them to the correct underlying causes.**

Analyzing mutational signatures requires a pristine dataset of only the mutations that have arisen in the tumor, which is why a robust filtering pipeline to identify and remove all germline variants is a mandatory first step.

Types of mutations



Types of mutations



Single base substitutions (SBS)

SBS-6 classification (base substitution)

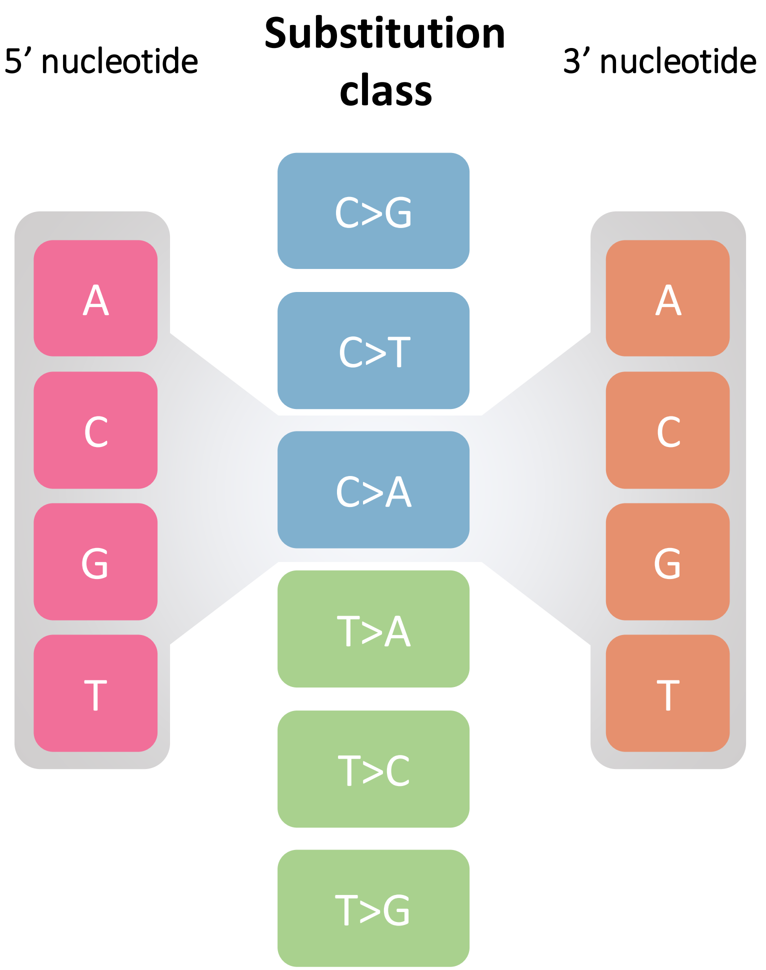
Substitution class

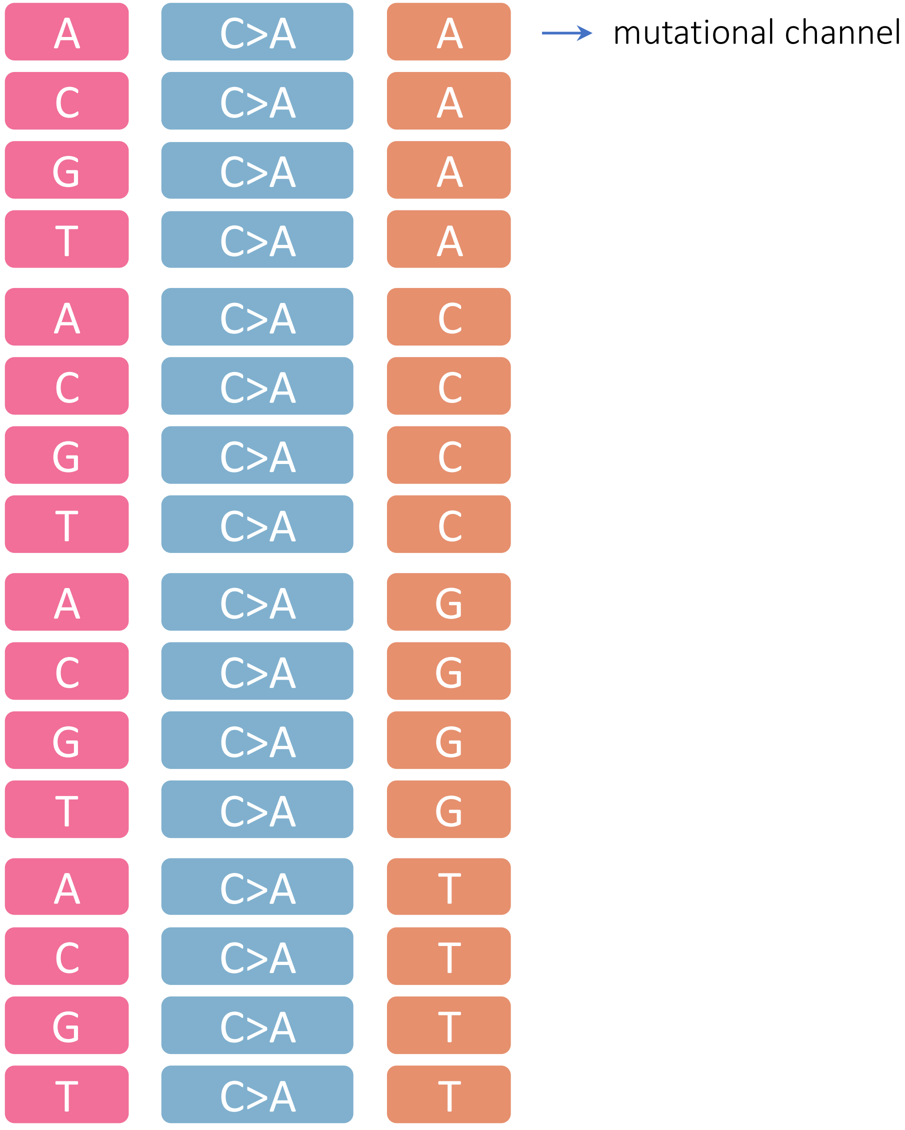
C>G	G>C
C>T	G>A
C>A	G>T
T>A	A>T
T>C	A>G
T>G	A>C

- single base substitutions can be described using only the mutated base-pair
- 6 possible mutational channels known as **SBS-6 classification**

The 96 trinucleotide context | SBS - 96

trinucleotide context (± 1 bp context)

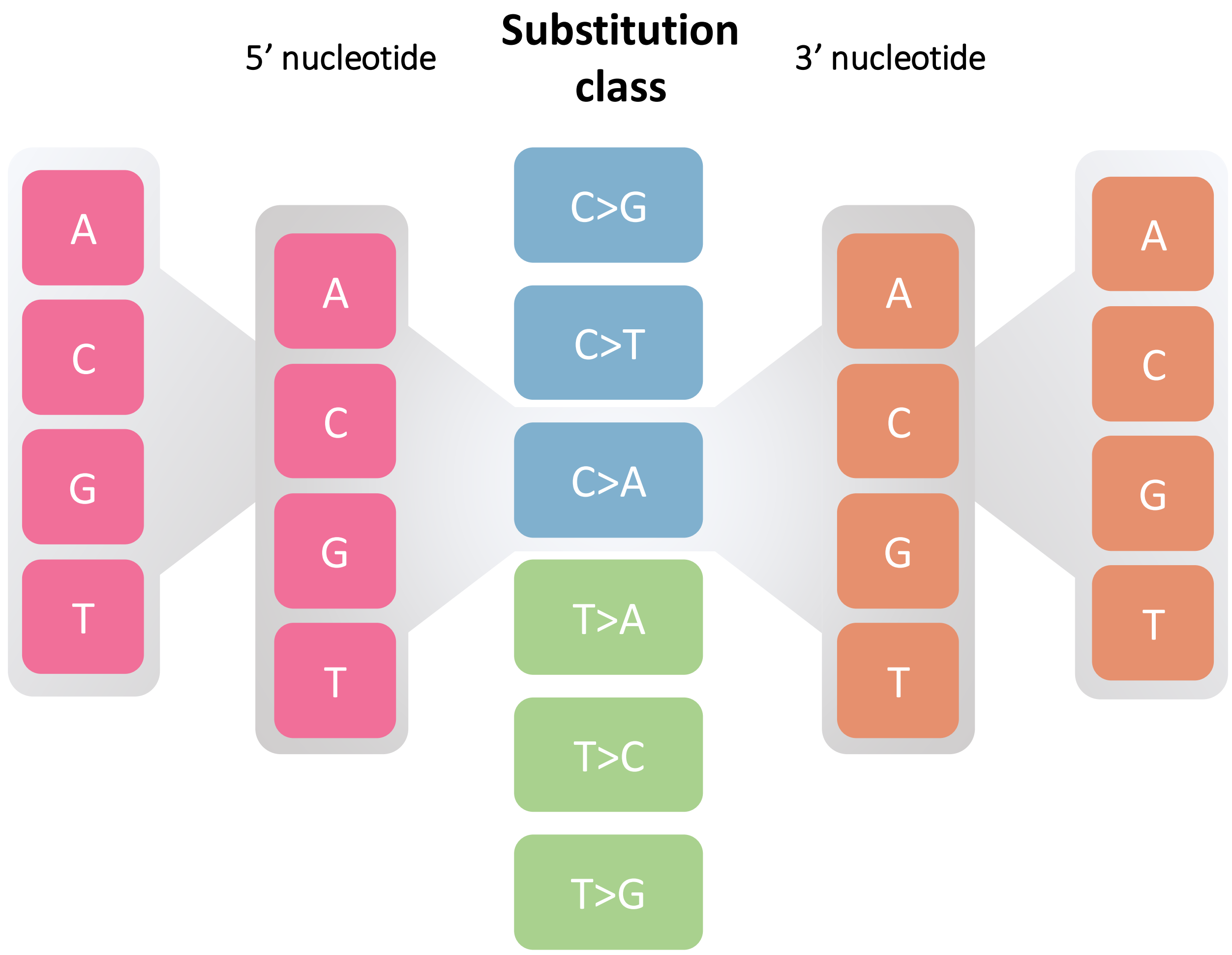




Example C>A 16 mutational channels

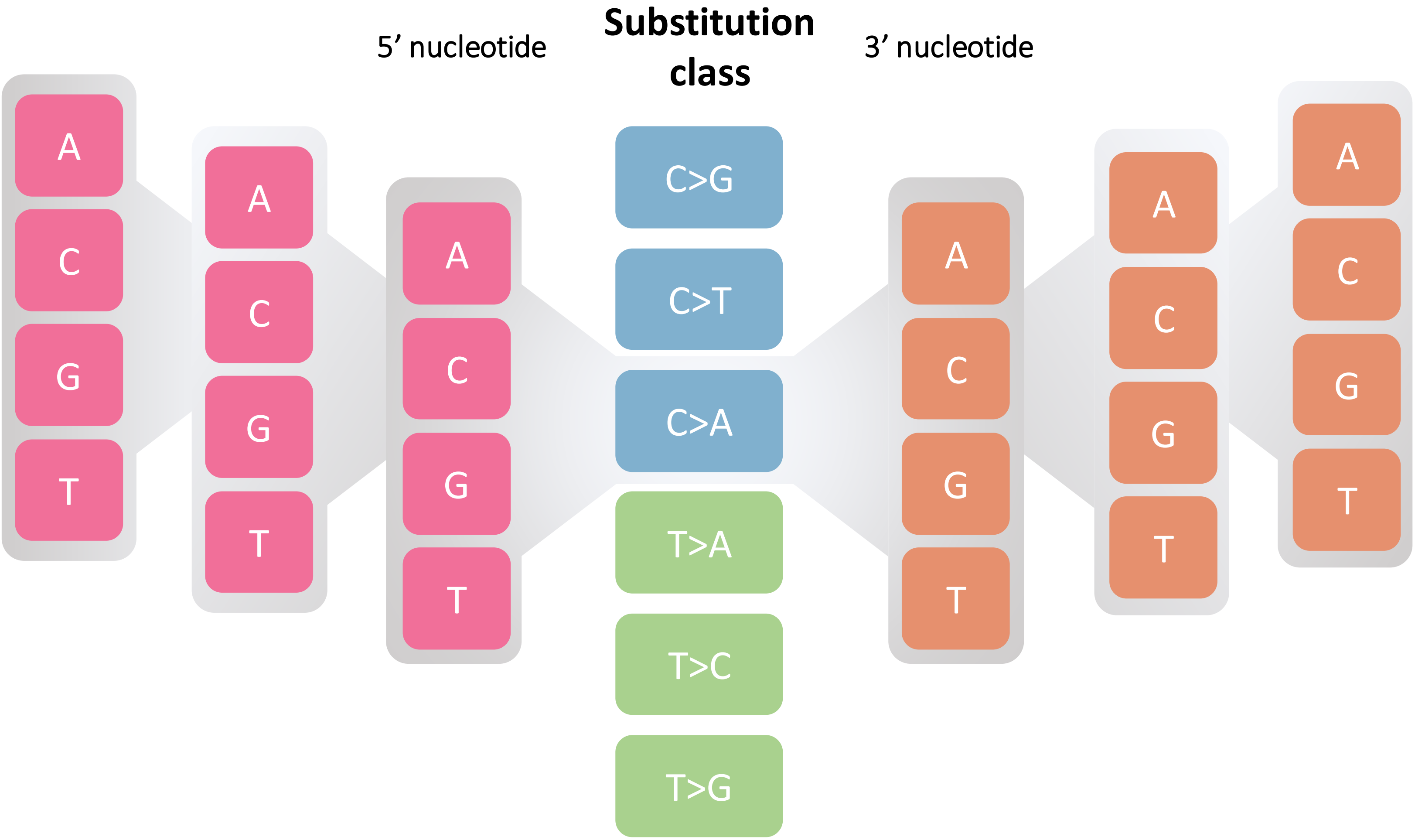
SBS - 1536

5-nucleotide context (± 2 bp context)



SBS - 24576

7-nucleotide context (± 3 bp context)



SBS - 24

extends SBS-6

SBS-96 → SBS-384

SBS-1536 → SBS-6144

Substitution class

- C>G
- C>T
- C>A
- T>A
- T>C
- T>G

C>A:

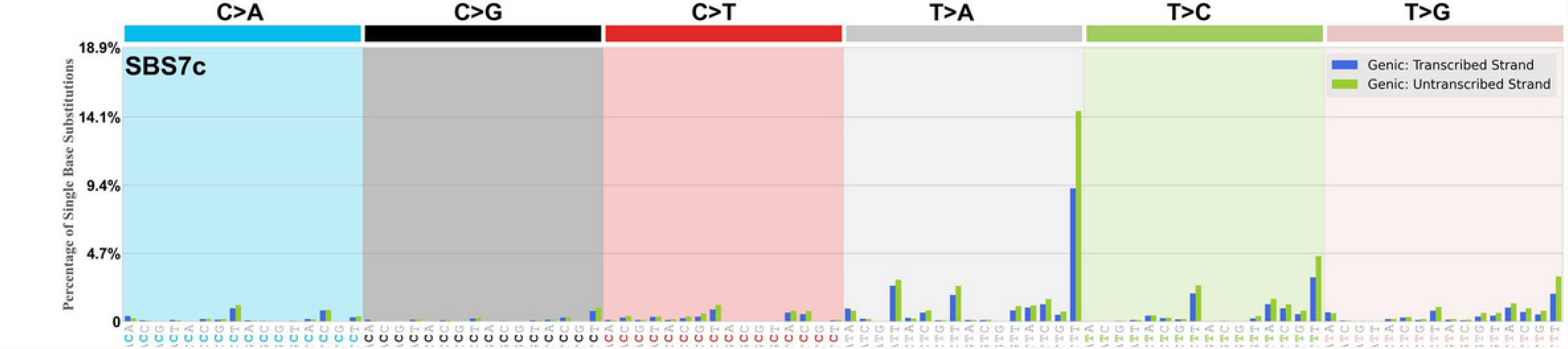
I. non-transcribed/intergenic regions

II. genic regions, occurring on the transcribed strand

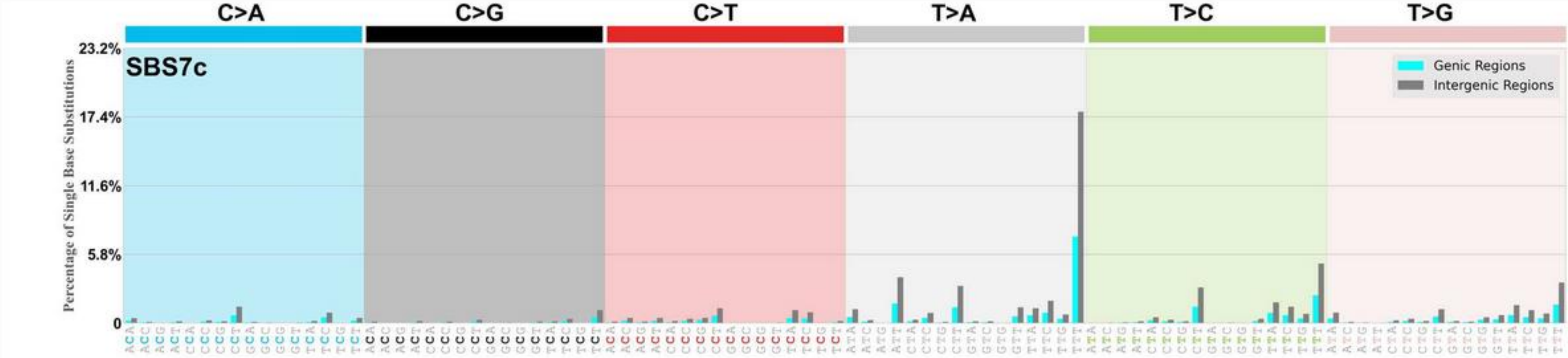
III. genic regions, occurring on the untranscribed strand

IV. genic regions, occurring in regions of bi-directional transcription

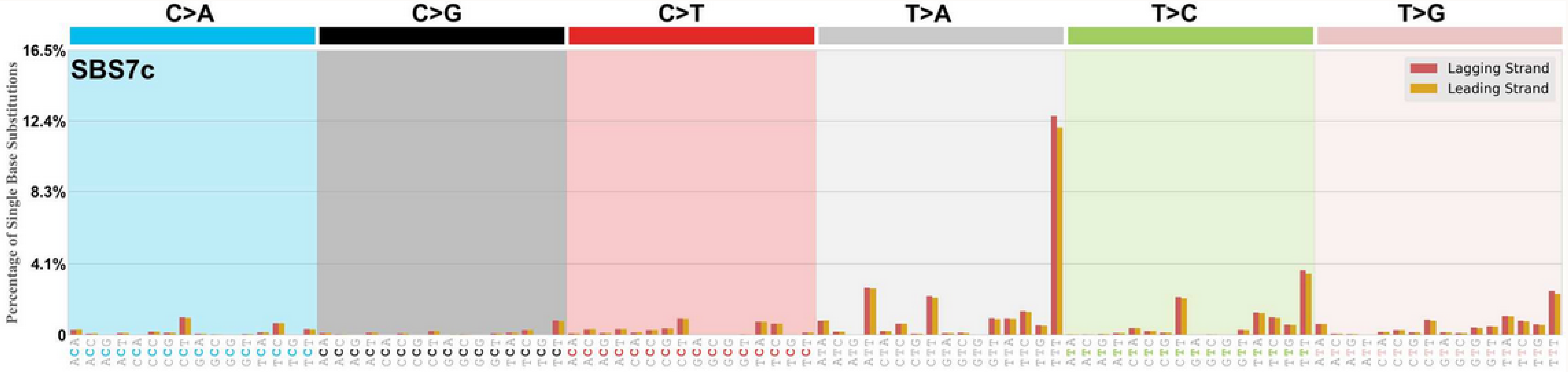
Transcriptional strand asymmetry



Genic and intergenic regions



Replicational strand asymmetry



SBS – 24 : transcriptional strand bias

1. Genic Regions, Untranscribed Strand

These are mutations that occur within the boundaries of a gene on the untranscribed strand. This is the complementary DNA strand that isn't being directly read by the RNA polymerase.

- ***Significance****: The untranscribed strand is less exposed and is not a direct substrate for transcription-coupled repair. Therefore, comparing the mutation patterns between the transcribed and untranscribed strands can give you powerful clues about the specific DNA repair deficiencies at play. A difference in the number of mutations between these two strands is called transcriptional strand bias.*

2. Genic Regions, Bidirectional Transcription

This is a more specialized category for mutations that occur in a genomic region that is transcribed on both strands. This happens in certain parts of the genome where two genes are very close together and are transcribed in opposite directions, with their start sites near each other.

- ***Significance****: In these areas, both strands are subject to the processes and repair mechanisms of transcription. Mutations in these regions provide a unique insight into mutational processes that are influenced by transcription on both strands.*

SBS – 24 : transcriptional strand bias

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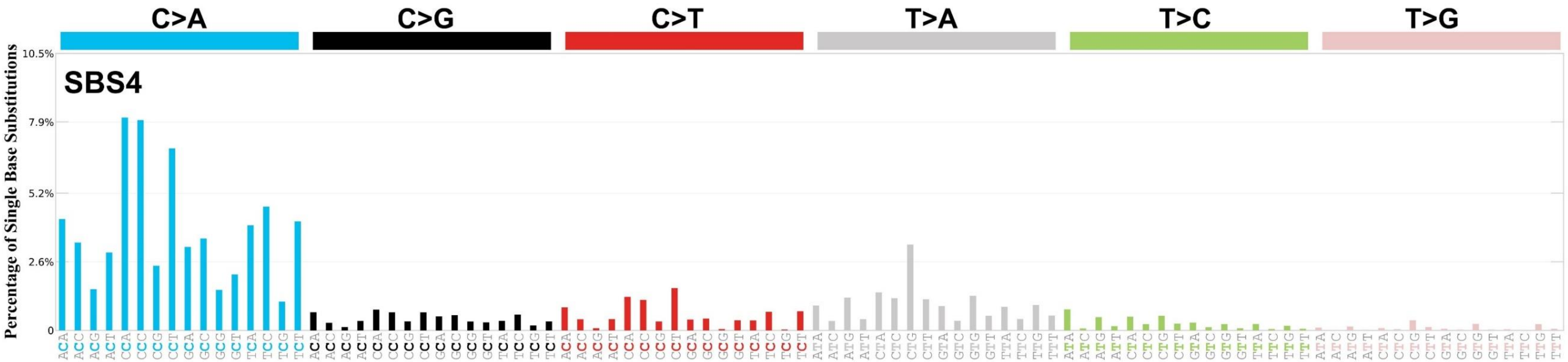
COSMIC Mutational Catalog (The Reference Library)

The Forensics Encyclopedia of Cancer Mutagens

- **What it is**: COSMIC is the gold-standard public repository for mutational signatures. Think of it as the forensics encyclopedia for cancer. It was meticulously built by experts analyzing mutations from thousands of tumors worldwide.
- **What it contains**: It contains the profiles (patterns) of all known mutational causes—the "smoking guns" of cancer. These profiles are called Single Base Substitution (SBS) signatures, such as SBS7 (the signature of UV light) and SBS4 (the signature of tobacco smoke).
- **Core Question**: When we use COSMIC, we are asking a simple but powerful question about a patient's tumor: "Does the pattern of mutations in this tumor match any of the known causes listed in the COSMIC encyclopedia?"

Single Base Substitution (SBS) Signatures

SBS 4 : tobacco smoking signature



Proposed aetiology

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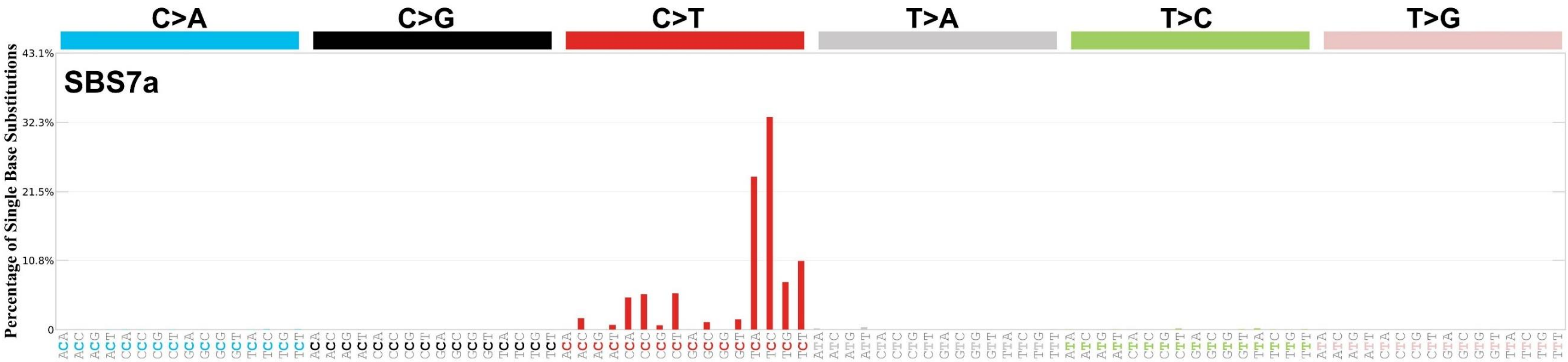
https://cancer.sanger.ac.uk/cosmic

SBS 7a : UV light exposure signature



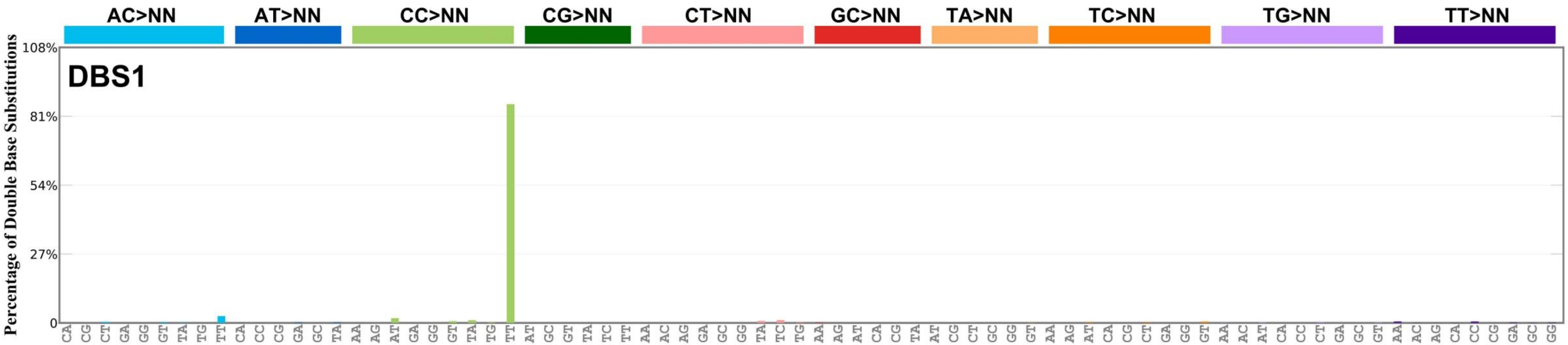
Proposed aetiology

SBS7a/SBS7b/SBS7c/SBS7d are found in cancers of the skin from sun exposed areas and are thus likely to be due to exposure to **ultraviolet light**. SBS7a may possibly be the consequence of just one of the two major known UV photoproducts, cyclobutane pyrimidine dimers or 6-4 photoproducts. However, there is currently no evidence for this hypothesis and it is unclear which of these photoproducts may be responsible for SBS7a.

https://cancer.sanger.ac.uk/cosmic

Doublet Base Substitution (DBS) Signatures

Doublet Base Substitution (DBS) Signatures



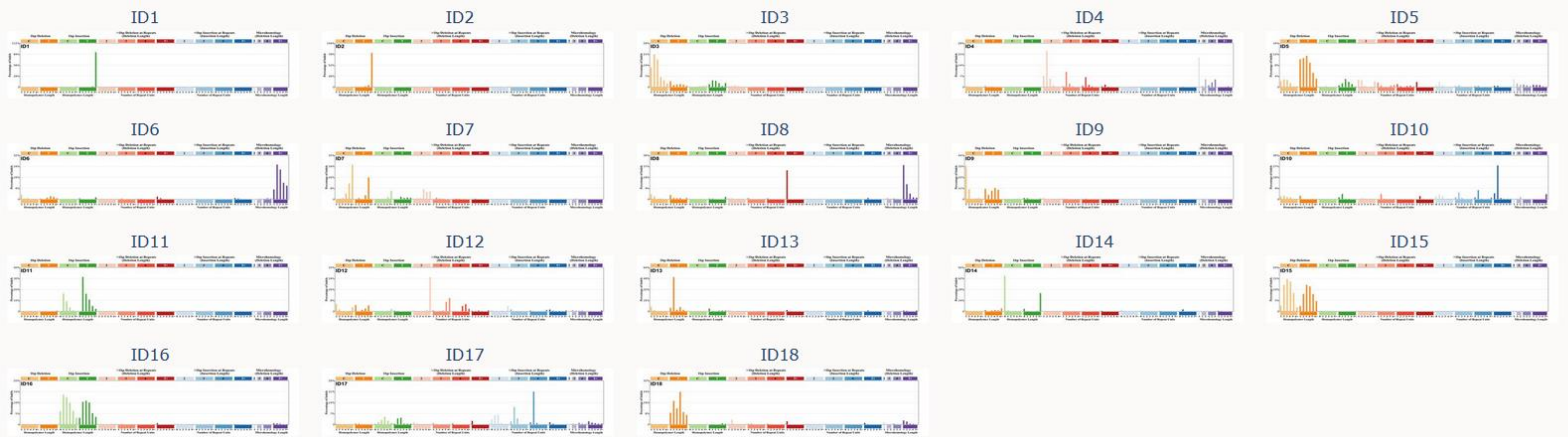
Proposed aetiology

Exposure to ultraviolet light.

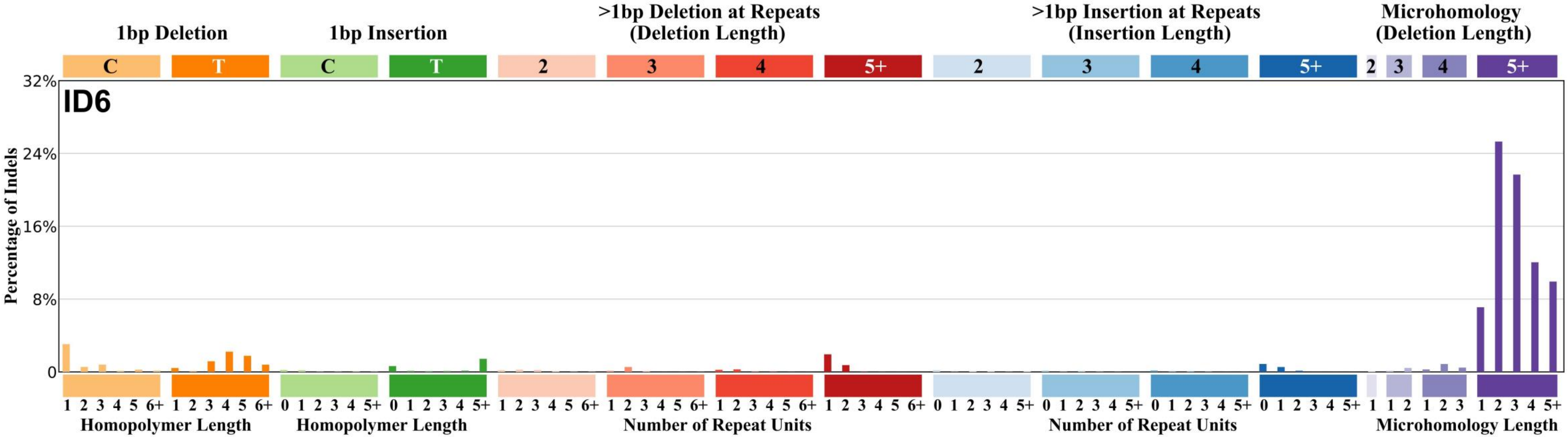
Comments

DBS1 exhibits transcriptional strand bias with more CC>TT mutations than GG>AA on the untranscribed strands of genes indicative of damage to cytosine and repair by transcription coupled nucleotide excision repair.

Small insertions and deletions (ID) Signatures



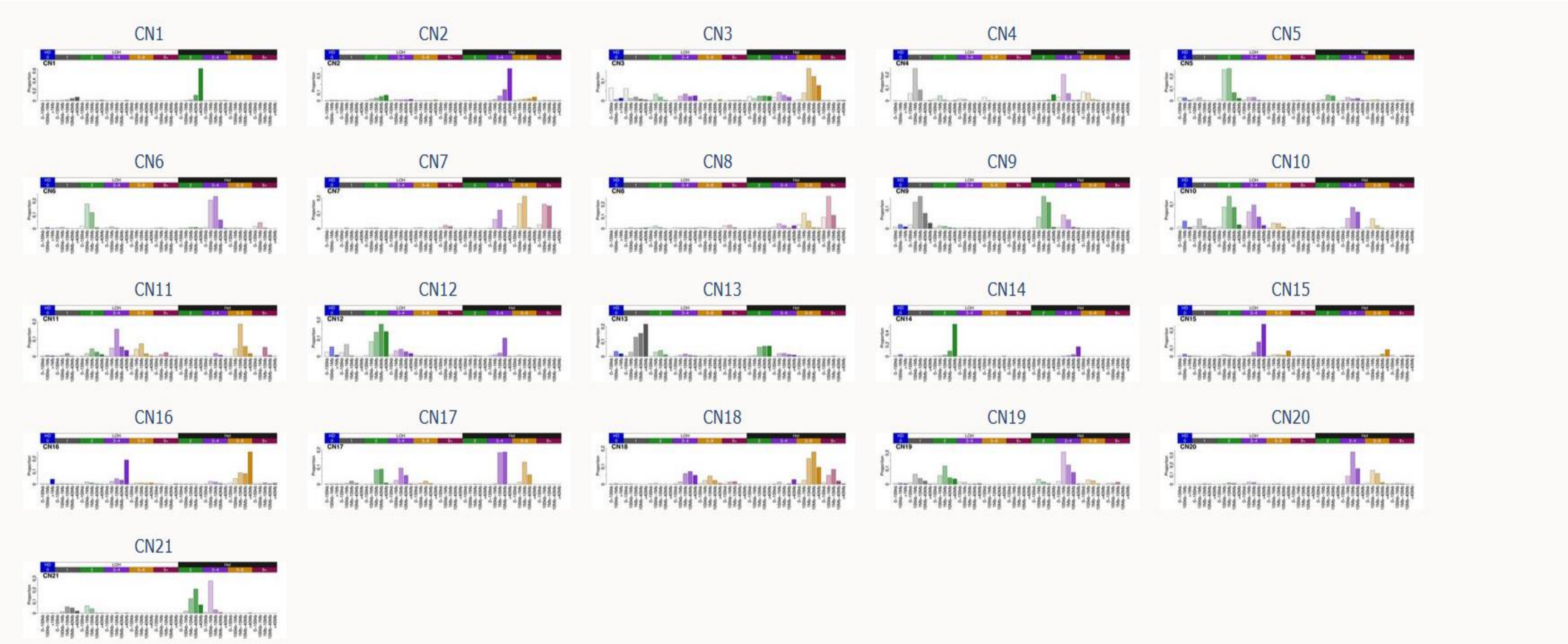
Small insertions and deletions (ID) Signatures



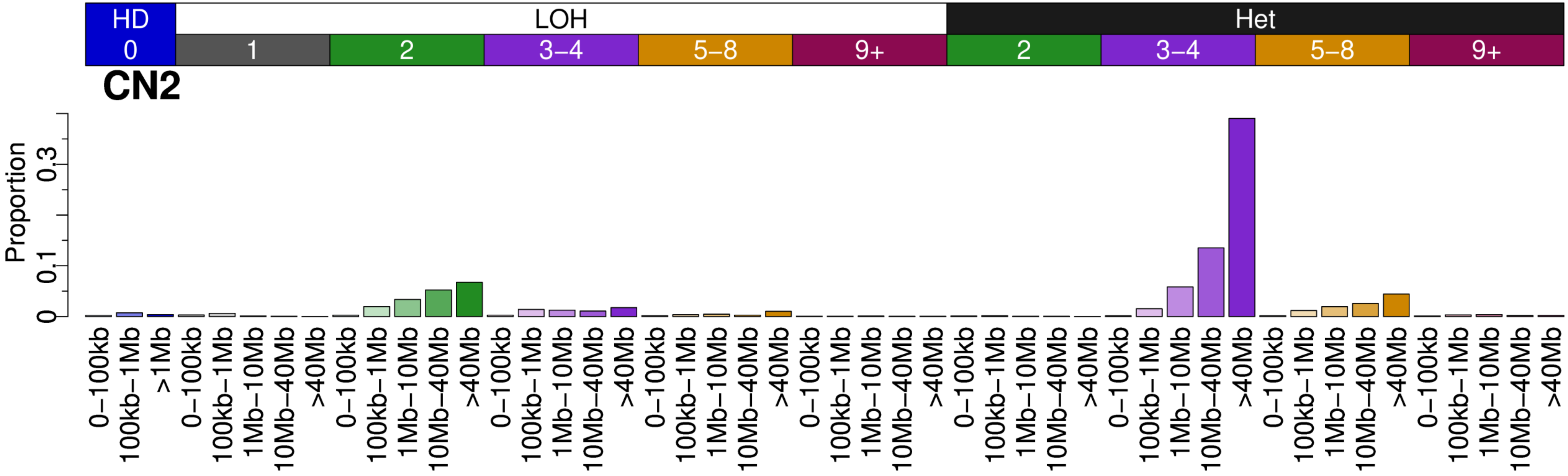
Proposed aetiology

Defective homologous recombination-based DNA damage repair, often due to inactivating BRCA1 or BRCA2 mutations, leading to non-homologous DNA end-joining activity.

Copy Number Variations (CN) Signatures



Copy Number Variations (CN) Signatures



CN2 is primarily a signature of a tetraploid genome; this has been verified through associations with once-genome-doubled samples, simulations of genome doubling, and artificial genome doubling of signature CN1. Note that due to the copy number class that most strongly defines this signature (TCN 3-4), a triploid cancer may be assigned this signature.
However, if CN2 is observed at <<100% proportion of segments, it may instead indicate aneuploidy, with gained segments (on a diploid background) being assigned CN2, and unaltered segments being assigned CN1 or lost segments being assigned CN9 or CN13, depending on size and genomic background.

Notes

CN2 is associated with a poor disease specific survival in thymomas.

De Novo Catalog (Building Your Own Library)

- A de novo catalog is a set of mutational signatures that you extract directly from your own dataset without any prior knowledge. You're creating new "fingerprints" based on the patterns present in your specific cohort of samples.
- Pure discovery. You're not looking for a match to a known pattern; you're letting the data speak for itself to find new, previously uncharacterized mutational processes that might be unique to your study population or cancer type.
- This as an unsupervised learning approach. You're performing NMF on your mutation matrix (V) to simultaneously solve for both the signature matrix (W) and the contribution matrix (H). The algorithm finds the optimal number of signatures and their patterns based on the data provided.

Key advantage: Discovery. This method can identify novel signatures that might not yet be in the COSMIC database.

When to Use Each Approach

- **Start with COSMIC**: *Always begin your analysis by fitting your data to the COSMIC catalog. It's the standard first step and will tell you if your tumors are driven by known, common mutational processes.*
- **Use De Novo for Discovery**: *If your data doesn't fit the COSMIC signatures well, or if you're working with a new or rare cancer type, a de novo extraction is appropriate. The signatures you discover can then be compared to the COSMIC catalog to see if they are similar to any known signatures or if they are truly novel.*

Methods for Extraction and Attribution

- **Brief intro to tools**
- **Matrix decomposition methods**
  - Non-negative matrix factorization (NMF).
- **Attribution methods**
  - Fitting known COSMIC signatures to new samples.
- **Practical considerations**
  - Sequencing depth, tumor purity, mutational burden.
  - Noise and overfitting.

Brief intro to tools

Table 1. Overview of bioinformatics tools for *de novo* extraction of mutational signatures

Tool name	Input	Platform	Factorization method	Factorization engine	GPU	Manual selection	Automatic selection	Automatic algorithm	Mutational catalog support	Plotting support	COSMIC comparison
EMu[20]	matrix	C++	EM	original implementation[20]	no	yes	yes[a]	BIC[21]	SBS-96	no	no
Maftools[22]	matrix, MAF	R-Bioconductor	NMF	NMF R package[23]	no	yes	no	–	SBS-96	SBS-96	1 to 1
Mutational Patterns[24]	matrix, VCF	R-Bioconductor	NMF	NMF R package[23]	no	yes	no	–	SBS-96, SBS-192	SBS-96, SBS-192	1 to 1
MutSignatures[25]	matrix, VCF, MAF	R	NMF	Brunet et al.[26]	no	no	no	–	SBS-96	SBS-96	1 to 1
MutSpec[27]	matrix, VCF, custom	Galaxy, Perl, R	NMF	NMF R package[23]	no	yes	no	–	SBS-96, SBS-192	SBS-96, SBS-192	1 to 1
SigFit[28]	matrix	R	Bayesian inference	Stan R package[29]	no	yes	yes[a]	Elbow method[30]	SBS-96	SBS-96, SBS-192	1 to 1

Islam et al., 2022, Cell Genomics 2, 100179

Brief intro to tools

Table 1. Overview of bioinformatics tools for *de novo* extraction of mutational signatures

Tool name	Input	Platform	Factorization method	Factorization engine	GPU	Manual selection	Automatic selection	Automatic algorithm	Mutational catalog support	Plotting support	COSMIC comparison
SigMiner[31]	matrix, MAF	R	(automatic) Bayesian NMF, (manual) NMF	(automatic) Signature Analyzer implementation,[32] (manual) NMF R package[23]	no	yes[a]	yes	ARD[33]	SBS-96, DBS-78, ID-83	generic	1 to 1
Signature Analyzer[32,34]	matrix, MAF	R (CPU),[18] Python (GPU)[19]	Bayesian NMF	original implementation[32,34]	yes	no	yes	ARD[33]	SBS-96, DBS-78, ID-83	SBS-96, DBS-78, ID-83	1 to 1
Signature ToolsLib[35]	matrix, VCF, custom	R	NMF	NMF R package[23]	no	yes	no	–	SBS-96, DBS-78, ID-83, SV-32	SBS-96, SV-32, generic	1 to 2
SigneR[36]	matrix, VCF	R-Bioconductor, C++	Bayesian NMF	original implementation[36]	no	yes	yes[a]	BIC[21]	SBS-96	SBS-96	no
SigProfiler Extractor	matrix, VCF, MAF, custom	Python, R wrapper	NMF	(current work) original implementation	yes	yes	yes[a]	NMFk[37]	SBS-96, DBS-78, ID-83, CN-48, others,[15] any	SBS-96, DBS-78, ID-83, CN-48, SV-32, others,[15] generic	1 to many

Islam et al., 2022, Cell Genomics 2, 100179

Brief intro to tools

Table 1. ***Continued***

Tool name	Input	Platform	Factorization method	Factorization engine	GPU	Manual selection	Automatic selection	Automatic algorithm	Mutational catalog support	Plotting support	COSMIC comparison
SigProfiler_PCAWG[12]	matrix, VCF, MAF, custom	Python, MATLAB	NMF	Brunet et al.[26]	no	yes	no	–	SBS-96, DBS-78, ID-83, others,[15] any	SBS-96, DBS-78, ID-83	no
Somatic Signatures[38]	matrix, VCF	R-Bioconductor	NMF, PCA	NMF R package[23] pcaMethods R package[39]	no	yes	no	–	SBS-96	SBS-96	no
Tensor Signatures[40]	VCF	Python	NTF	TensorFlow[41]	yes	yes	yes[a]	BIC[21]	tensor	SBS-96 with strand bias	no

Tools are ordered alphabetically. 1 to 1 refers to one *de novo* signature being matched with exactly one COSMIC signature; 1 to 2 refers to one *de novo* signature being matched with a combination of up to two COSMIC signatures; 1 to many refers to one *de novo* signature being matched with a combination of one or more COSMIC signatures. MAF, mutation annotation format; VCF, variant call format; EM, expectation maximization algorithm; NMF, nonnegative matrix factorization; PCA, principal component analysis; NTF, nonnegative tensor factorization; ARD, automatic relevance determination; BIC, Bayesian information criterion; COSMIC, catalog of somatic mutations in cancer; SBS, single base substitutions; DBS, doublet base substitutions; ID, small insertions and deletions; CN, copy number; SV, structural variants.

[a]The default approach for selecting the total number of signatures when a tool supports both manual and automatic selection.

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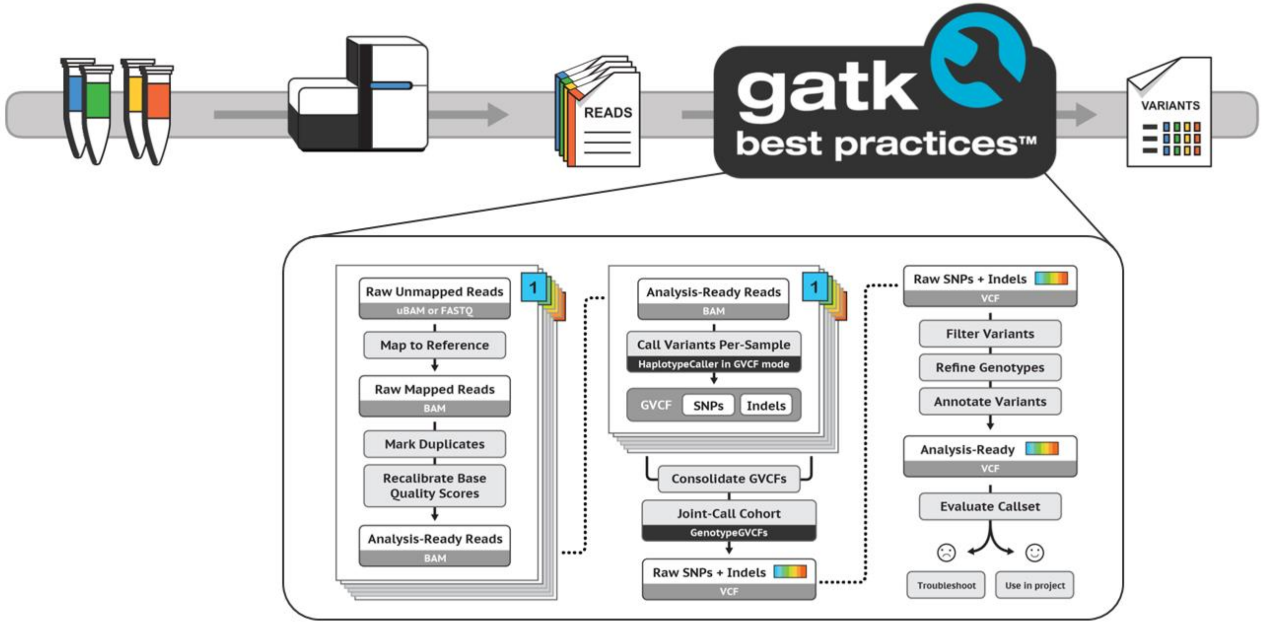
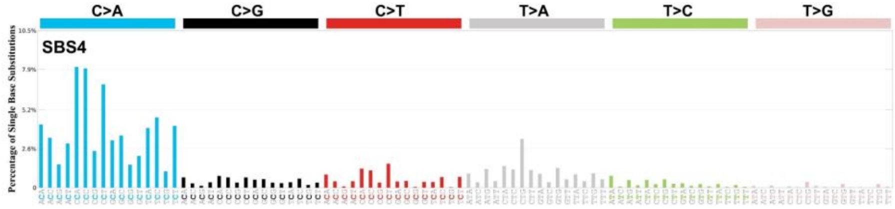
Somatic genomic mutations



Variant calling

Mutational signature inference

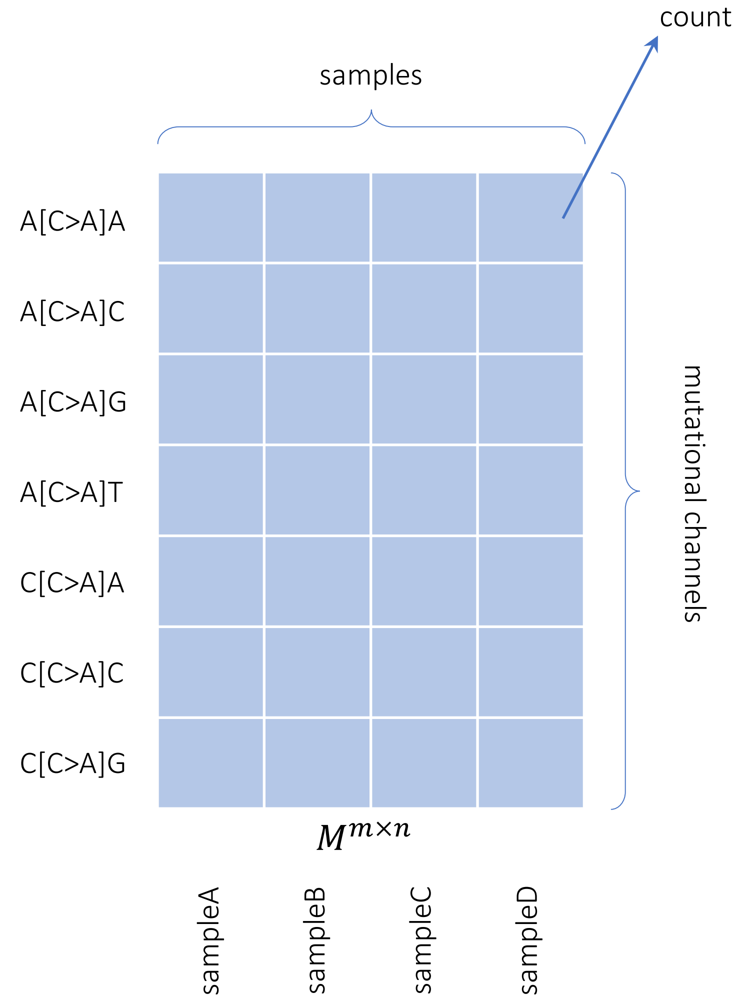


Transformation of somatic mutations into a matrix

somatic mutations in a cancer sample

```
Chromosome  Position    RefCall AltCall VAF Depth   CytoBand     GeneName
chr1    2488153 A   G   0.4406  202 1p36.32 TNFRSF14    SNV COSM4999217;CO
chr1    4367323 G   A   0.9966  290 1p36.32     SNV NA  0   NA  NA  0   14
chr1    8074334 C   T   0.5011  459 1p36.23 ERRFI1  SNV COSM3689848 1   NA
chr1    11169789    A   G   0.4684  316 1p36.22 MTOR    SNV NA  0   NA  NA
chr1    11181327    C   T   0.5641  234 1p36.22 MTOR    SNV COSM3996743 3
chr1    11190646    G   A   0.5659  334 1p36.22 MTOR    SNV COSM3996746 2
chr1    11205058    C   T   0.5804  367 1p36.22 MTOR    SNV COSM4142146 3
chr1    16256007    T   C   1   386 1p36.13 SPEN    SNV COSM4142934 4   NA
chr1    16259813    A   G   1   294 1p36.13 SPEN    SNV COSM4142938 3   NA
chr1    16260803    G   A   0.5018  277 1p36.13 SPEN    SNV     0   NA  NA
chr1    18957546    G   A   0.543   221 1p36.13 PAX7    SNV NA  0   NA  NA
chr1    19018405    G   A   0.5121  207 1p36.13 PAX7    SNV COSM3750707;CO
chr1    19018432    A   C   0.4781  228 1p36.13 PAX7    SNV COSM6351865;CO
chr1    19027239    A   G   0.3538  130 1p36.13 PAX7    SNV COSM424864;COS
chr1    19071752    T   C   1   251 1p36.13 PAX7    SNV NA  0   NA  NA  0
chr1    19072926    A   G   1   347 1p36.13 PAX7    SNV     0   NA  NA  0
chr1    19075225    A   G   1   343 1p36.13 PAX7    SNV NA  0   NA  NA  0
chr1    23885498    T   C   0.9973  371 1p36.12 ID3 SNV NA  0   NA  NA  20
chr1    40356155    A   G   0.4735  359 1p34.2      SNV NA  0   NA  NA  0
chr1    40363054    G   C   1   238 1p34.2  MYCL    SNV COSM3927628 1   NA
chr1    40364803    C   A   0.4777  224 1p34.2  MYCL    SNV NA  0   NA  NA
chr1    40366222    C   A   0.5227  308 1p34.2  MYCL    SNV NA  0   NA  NA
chr1    45797505    C   G   0.4784  255 1p34.1  MUTYH   SNV COSM3751252 21
chr1    45805566    G   C   1   94  1p34.1  MUTYH;TOE1  SNV NA  0   NA  NA
```

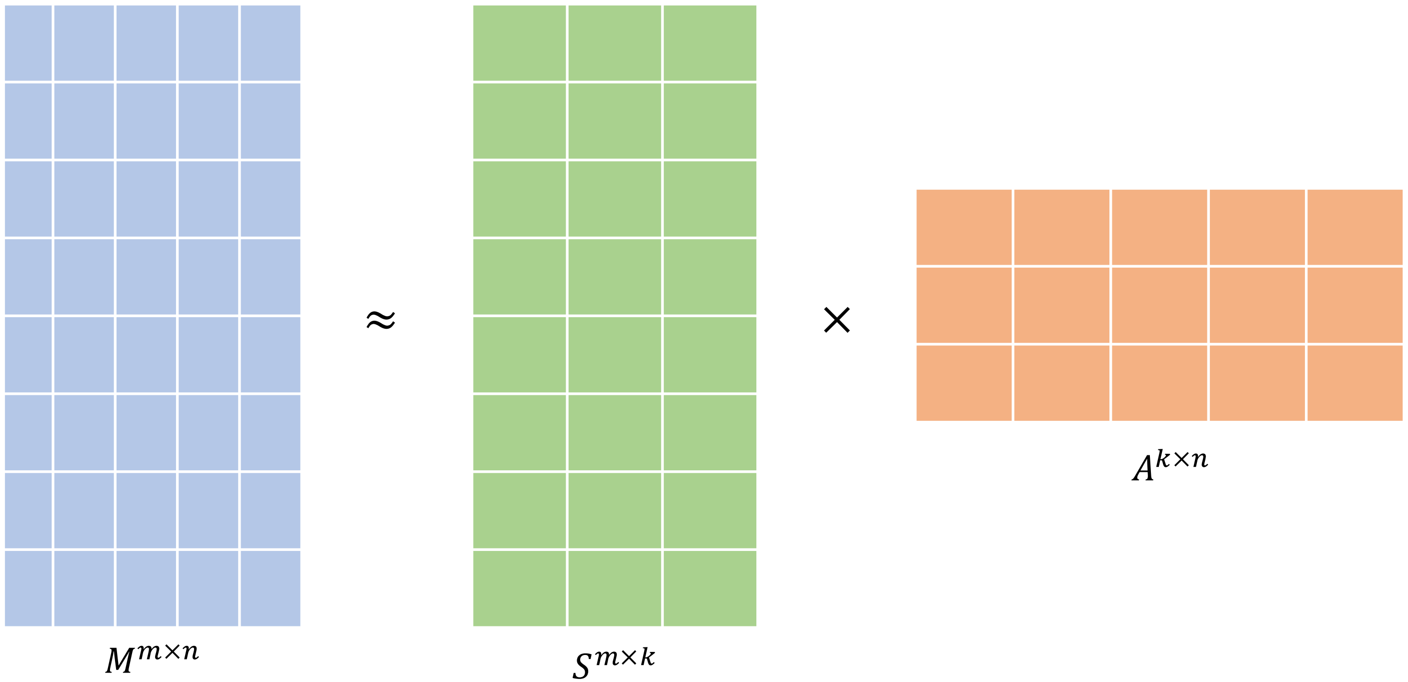
sampleA

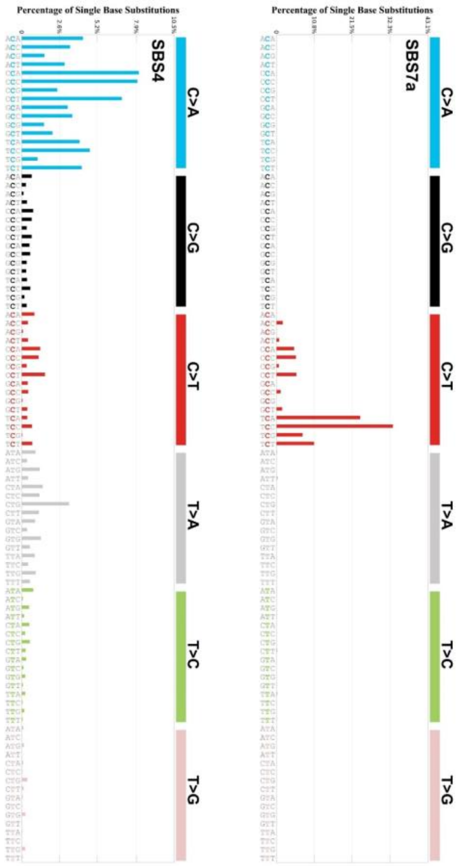


The value of each cell in the matrix, M, corresponds to the number of somatic mutations from a particular mutational channel in each sample

Nonnegative Matrix Factorization (NMF)

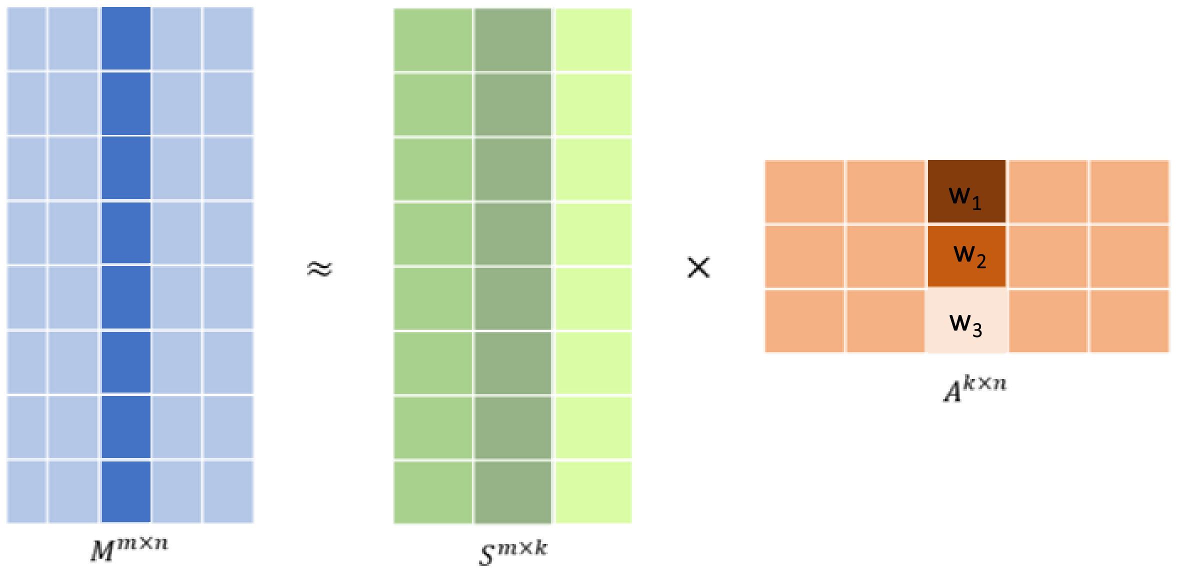
Note: the *S* and *A* matrices are not unique!

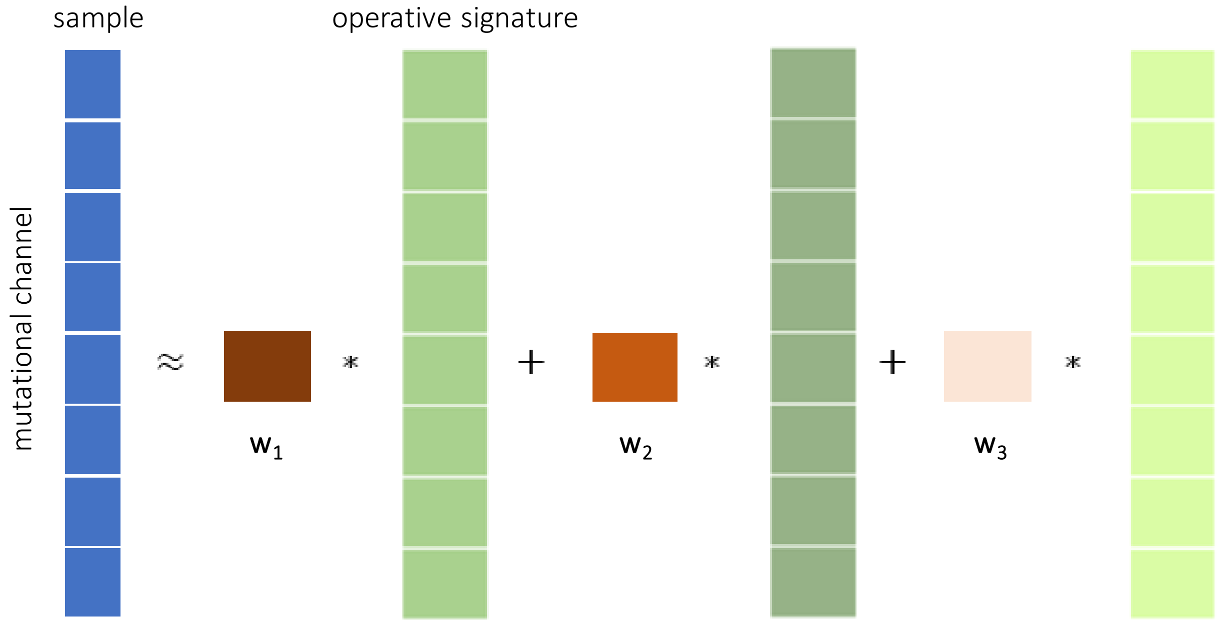




Note: the *S* and *A* matrices are not unique!

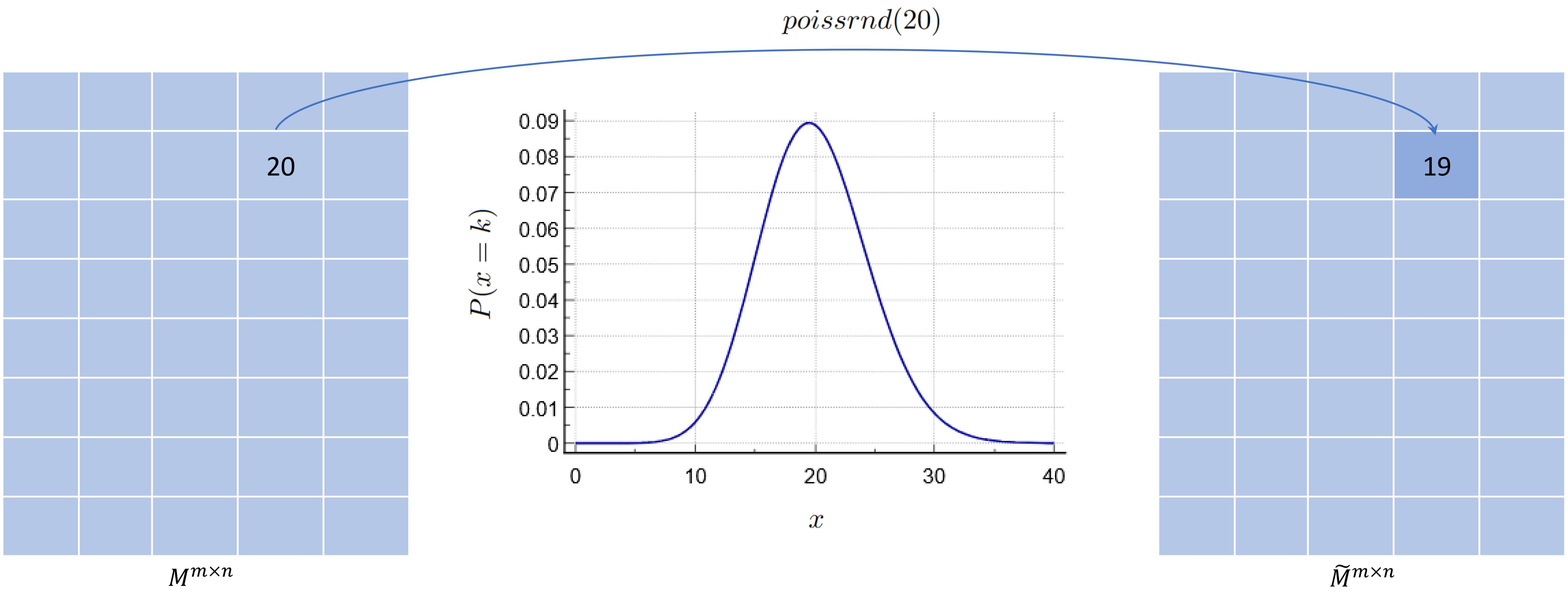
Nonnegative Matrix Factorization (NMF)





Resampling of the input mutational matrix

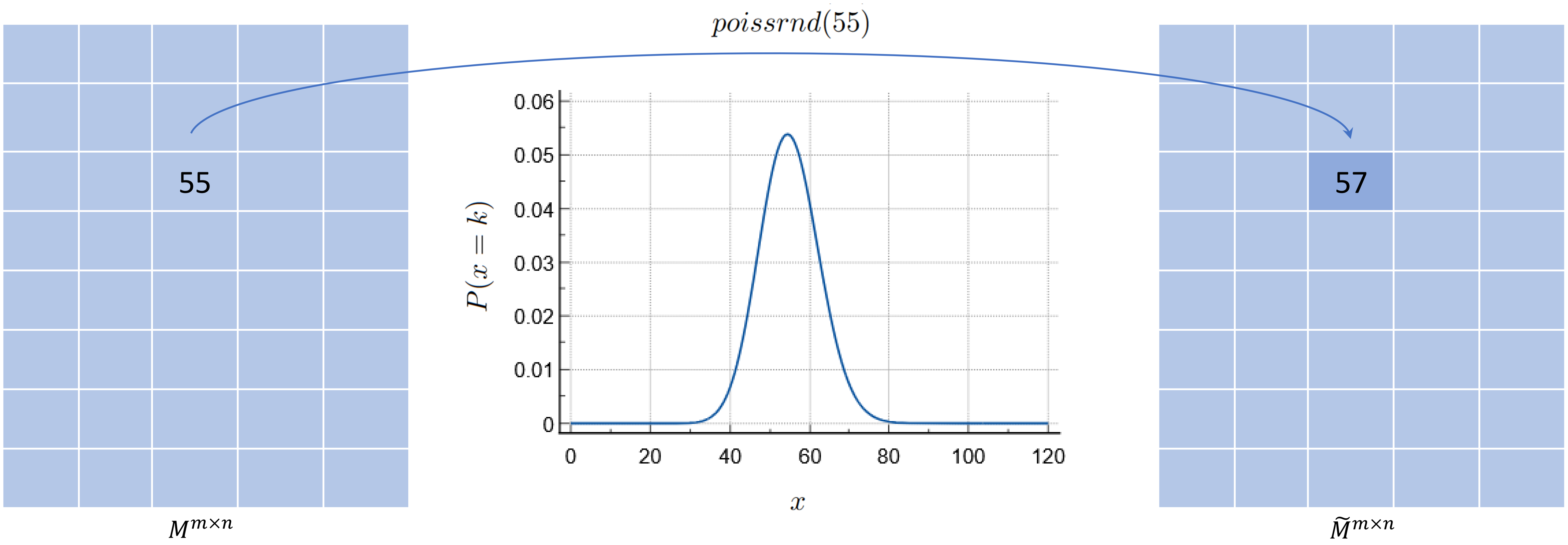
independent Poisson resampling of the original matrix for each replicate



The resampling is performed to ensure that Poisson fluctuations of the matrix do not impact the stability of the factorization results

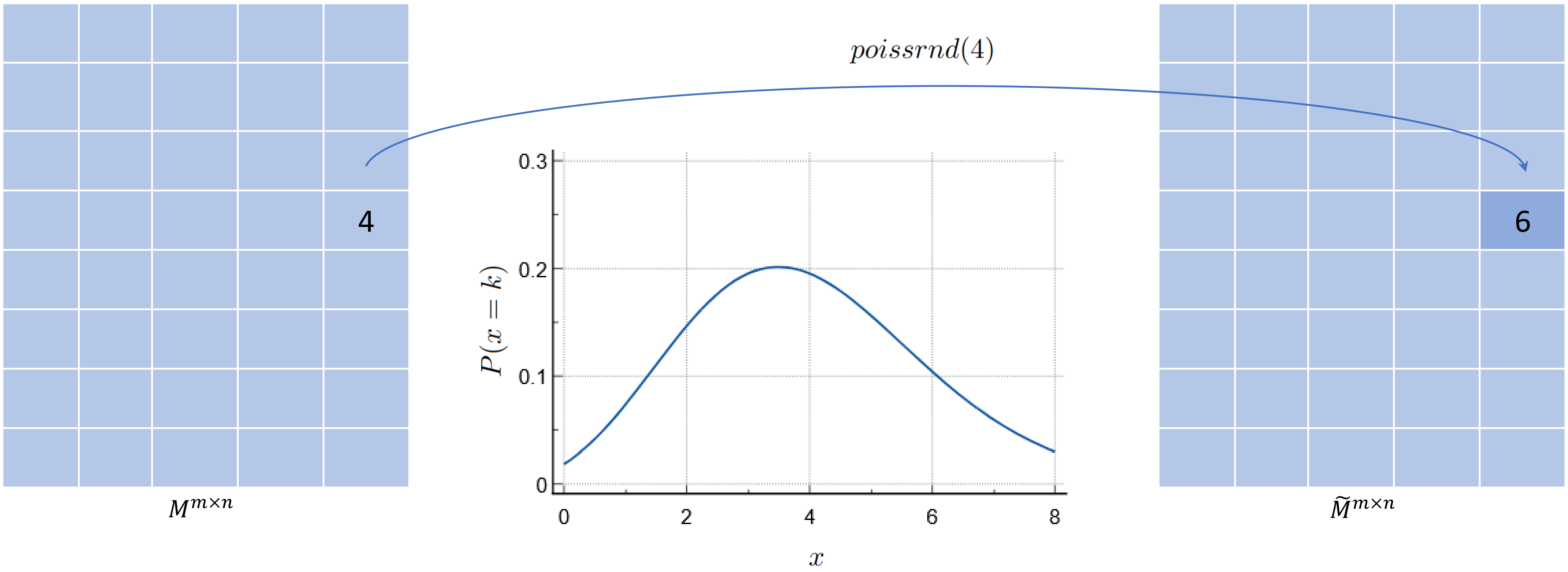
Resampling of the input mutational matrix

independent Poisson resampling of the original matrix for each replicate



Resampling of the input mutational matrix

independent Poisson resampling of the original matrix for each replicate

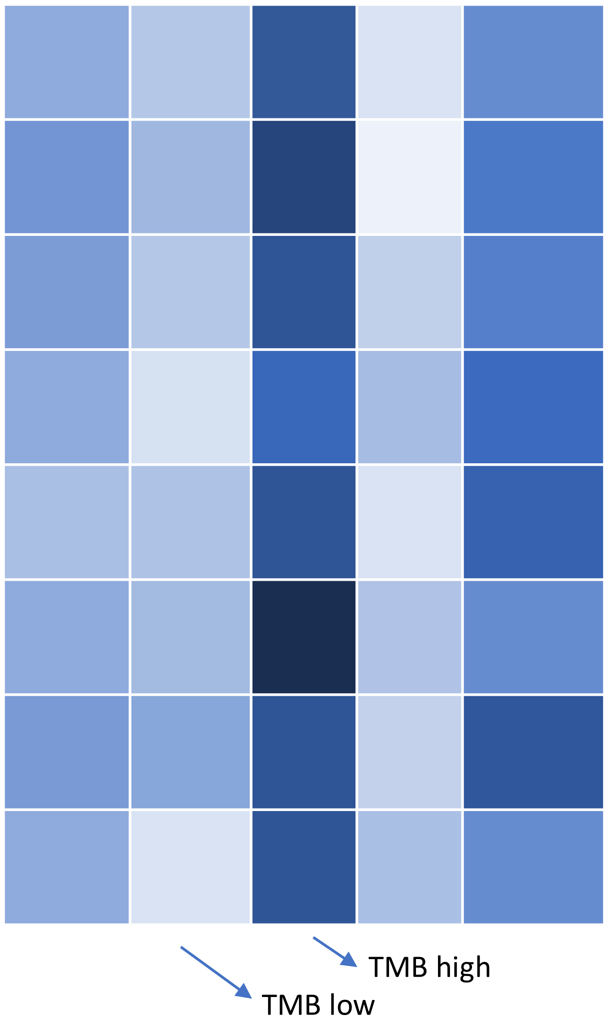


Resampling is a way to prove that the identified mutational signatures are real

Normalizing the resampled matrix

normalization to overcome potential skewing

$\widetilde{M}^{m \times n}$



- **Log2 normalization**

$$M(i,j) = M(i,j) \times \frac{\log_2\left(\sum_i M(i,j)\right)}{\sum_i M(i,j)}$$

- **100x normalization**

$$\text{if} \sum_i M(i,j) > 100 \times m$$

$$M(i,j) = \frac{M(i,j) \times 100 \times m}{\sum_i M(i,j)}$$

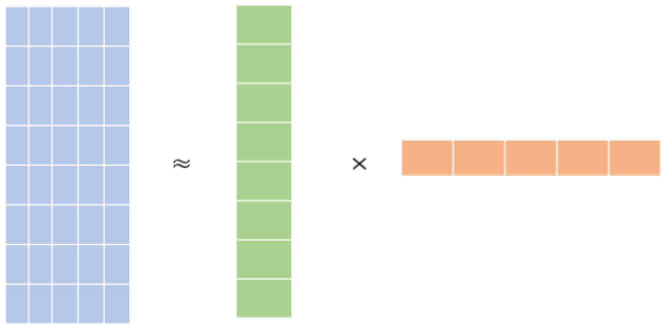
- **Gaussian mixture model (GMM) normalization**

- **No normalization**

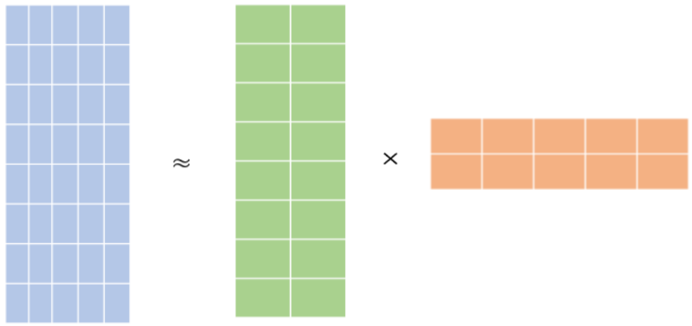
Module 3: Non-Negative Matrix factorization with replicates

rank

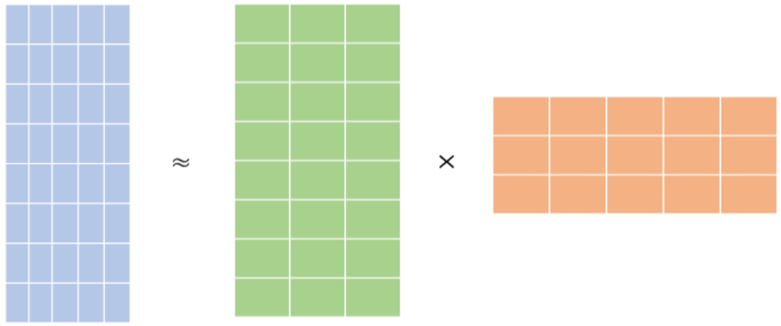
k = 1



k = 2



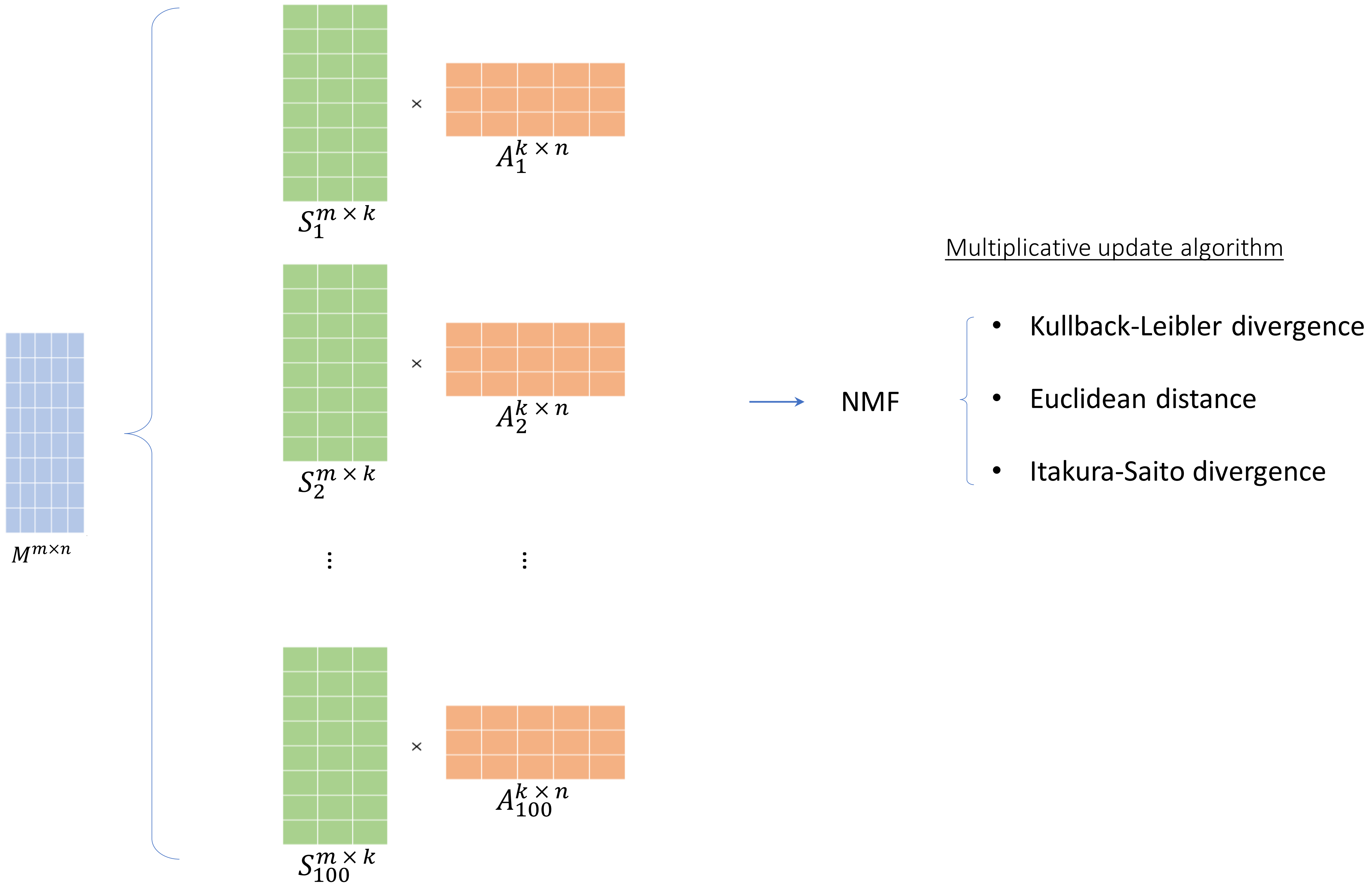
k = 3



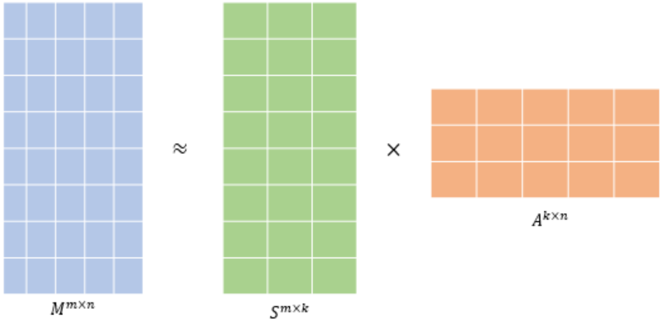
...

k = 25 ...

factorizes the matrix M with different ranks searching for an optimal solution between $k = 1$ and $k = 25$ mutational signatures (number of operative signatures)



$M^{m \times n}$
$S_1^{m \times k}$
$A_1^{k \times n}$
$S_2^{m \times k}$
$A_2^{k \times n}$
$S_{100}^{m \times k}$
$A_{100}^{k \times n}$
NMF
Multiplicative update algorithm
Kullback-Leibler divergence
Euclidean distance
Itakura-Saito divergence



Itakura-Saito divergence

The Itakura-Saito cost function:

$$D_{\mathrm{IS}}(\mathbf{X} \mid \mathbf{WH}) = \sum_{i,j}\left(\frac{\mathbf{X}_{ij}}{(\mathbf{WH})_{ij}} - \log\frac{\mathbf{X}_{ij}}{(\mathbf{WH})_{ij}} - 1\right)$$

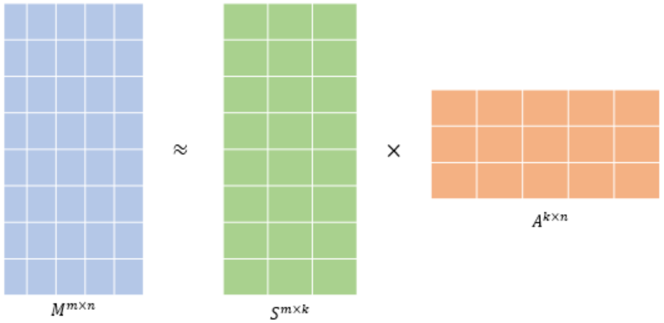
$M^{m\times n}$ $\quad S^{m\times k}$ $\quad A^{k\times n}$

$$\frac{\partial f(x)}{\partial x} = \frac{c}{x} - \log\frac{c}{x} - 1 = \frac{\partial}{\partial x}\left(\frac{c}{x} - \log c + \log x - 1\right) = -\frac{c}{x^2} + \frac{1}{x}$$

Deriving with respect to w_{ij}

$$\frac{\partial D_{\mathrm{IS}}(\mathbf{X},\mathbf{WH})}{\partial w_{ij}} = \sum_{m,n} x_{mn}\frac{\partial}{\partial w_{ij}}\frac{1}{(\mathbf{WH})_{mn}} + \sum_{m,n}\frac{\partial}{\partial w_{ij}}\log(\mathbf{WH})_{mn}$$

$$\nabla_{\mathbf{W}} D_{\mathrm{IS}}(\mathbf{X},\mathbf{WH}) = -\frac{\mathbf{X}}{(\mathbf{WH})^{\circ 2}}\mathbf{H}^{\intercal} + \frac{1}{\mathbf{WH}}\mathbf{H}^{\intercal}$$



Itakura-Saito divergence

The Itakura-Saito cost function:

$$D_{\mathrm{IS}}(\mathbf{X} \mid \mathbf{WH}) = \sum_{i,j} \left(\frac{\mathbf{X}_{ij}}{(\mathbf{WH})_{ij}} - \log \frac{\mathbf{X}_{ij}}{(\mathbf{WH})_{ij}} - 1 \right)$$

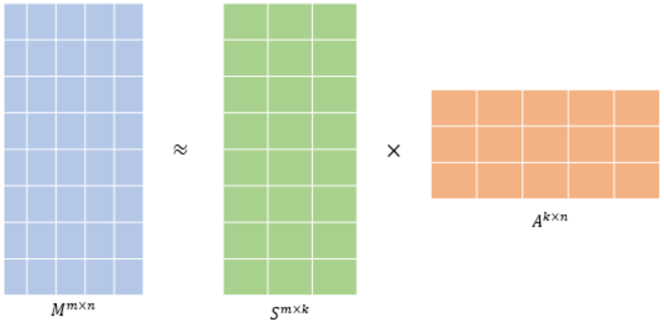
$M^{m\times n}$ $\quad S^{m\times k}$ $\quad A^{k\times n}$

Minimization problem

$$\frac{\partial f(x)}{\partial x} = \frac{c}{x} - \log\frac{c}{x} - 1 = \frac{\partial}{\partial x}\left(\frac{c}{x} - \log c + \log x - 1\right) = -\frac{c}{x^2} + \frac{1}{x}$$

Deriving with respect to h_{ij}

$$\nabla_{\mathbf{H}} D_{\mathrm{IS}}(\mathbf{X}, \mathbf{WH}) = -\mathbf{W}^{\intercal}\frac{\mathbf{X}}{(\mathbf{WH})^{\circ 2}} + \mathbf{W}^{\intercal}\frac{1}{\mathbf{WH}}$$



Itakura-Saito divergence

$$M^{m\times n} \approx S^{m\times k} \times A^{k\times n}$$

$$\mathbf{H} \leftarrow \mathbf{H} \circ \frac{\mathbf{W}^{\intercal} \frac{\mathbf{X}}{(\mathbf{WH})^{\circ 2}}}{\mathbf{W}^{\intercal} \frac{1}{\mathbf{WH}}}$$

↓

$$S^{m\times k}$$

$$\mathbf{W} \leftarrow \mathbf{W} \circ \frac{\frac{\mathbf{X}}{(\mathbf{WH})^{\circ 2}} \mathbf{H}^{\intercal}}{\frac{1}{\mathbf{WH}} \mathbf{H}^{\intercal}}$$

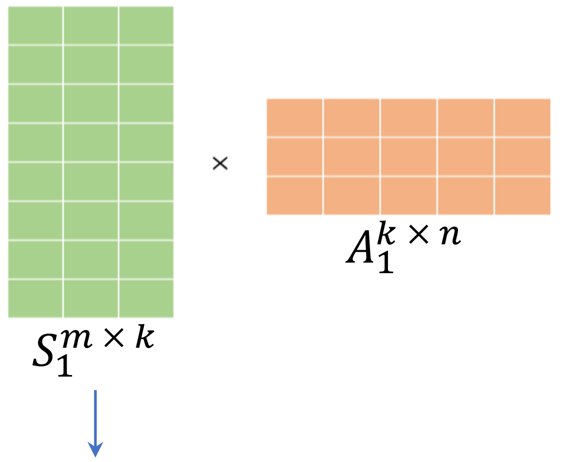
↓

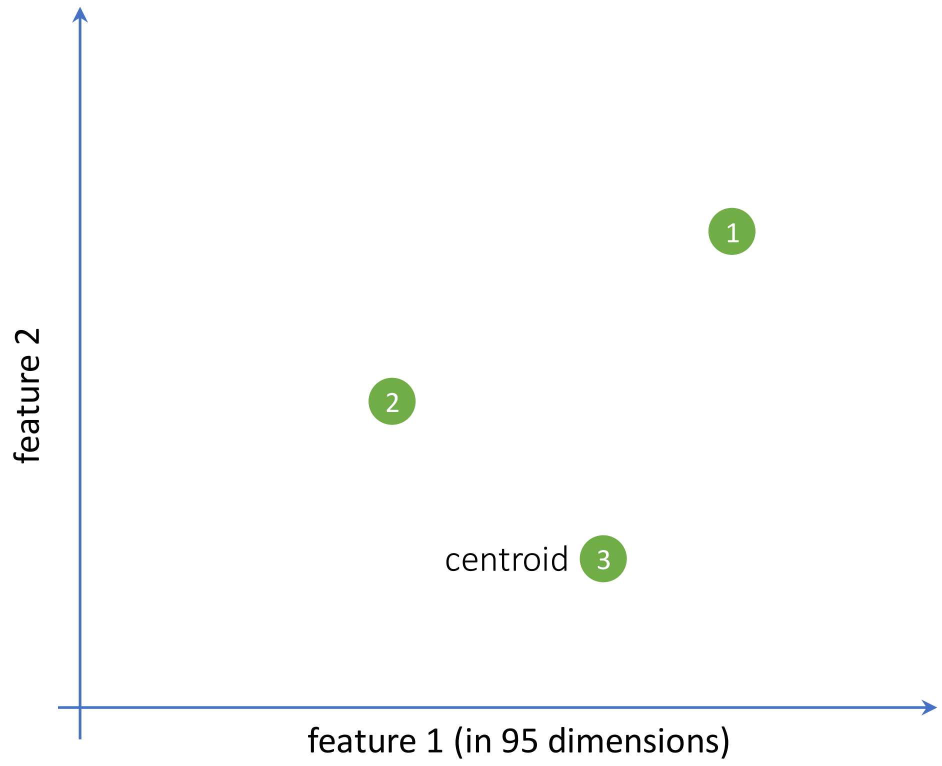
$$A^{k\times n}$$

10’000 – 1’000’000 NMF multiplicative update steps

Partition clustering

k = 3 (number of operative signatures)



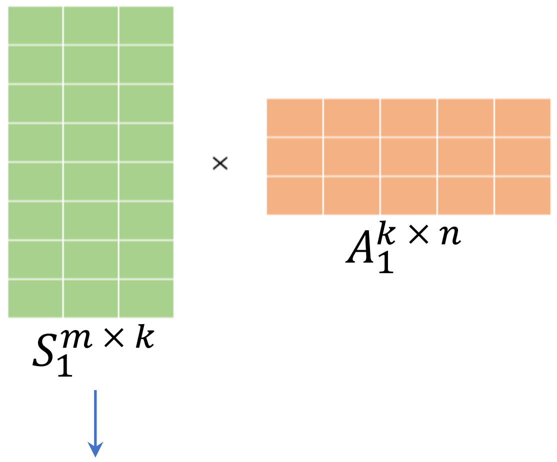


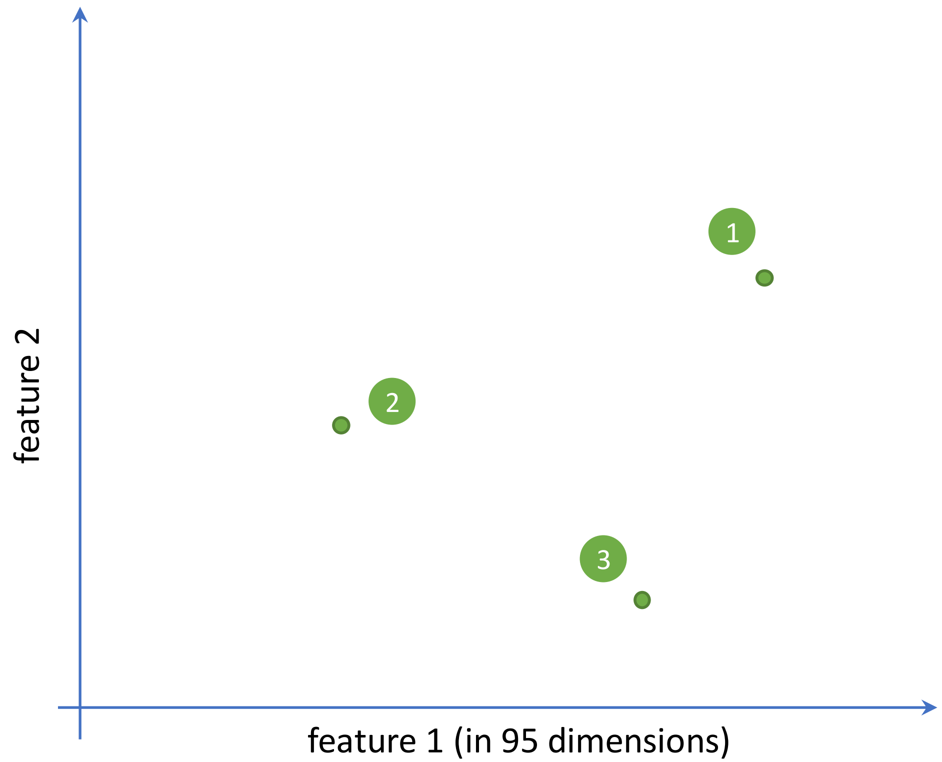
<u>Clustering algorithm</u>

1. ***k* clusters initialized randomly**
2. Each column of S_i assigned to a different centroid (in 96 dimensions)
3. Repeat for all 100 S_i matrices
4. After assigning all columns to a cluster, the centroid of each cluster is recalculated

Partition clustering

k = 3 (number of operative signatures)





<u>Clustering algorithm</u>

1. k clusters initialized randomly
2. **Each column of S_i assigned to a different centroid (in 96 dimensions)**
3. Repeat for all 100 S_i matrices
4. After assigning all columns to a cluster, the centroid of each cluster is recalculated

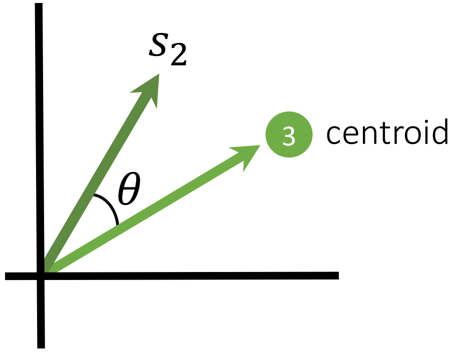
Partition clustering

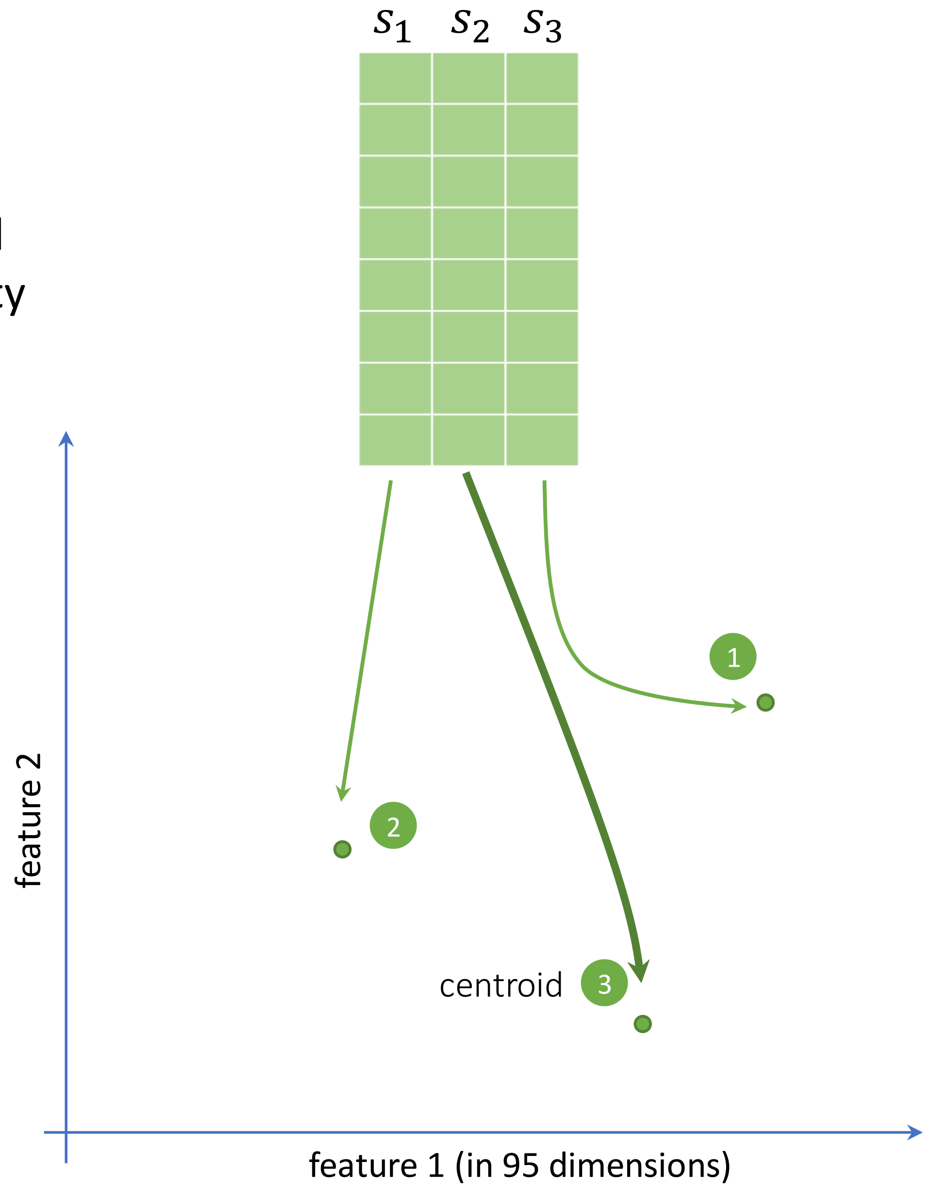
Each column of S_i assigned to a different centroid

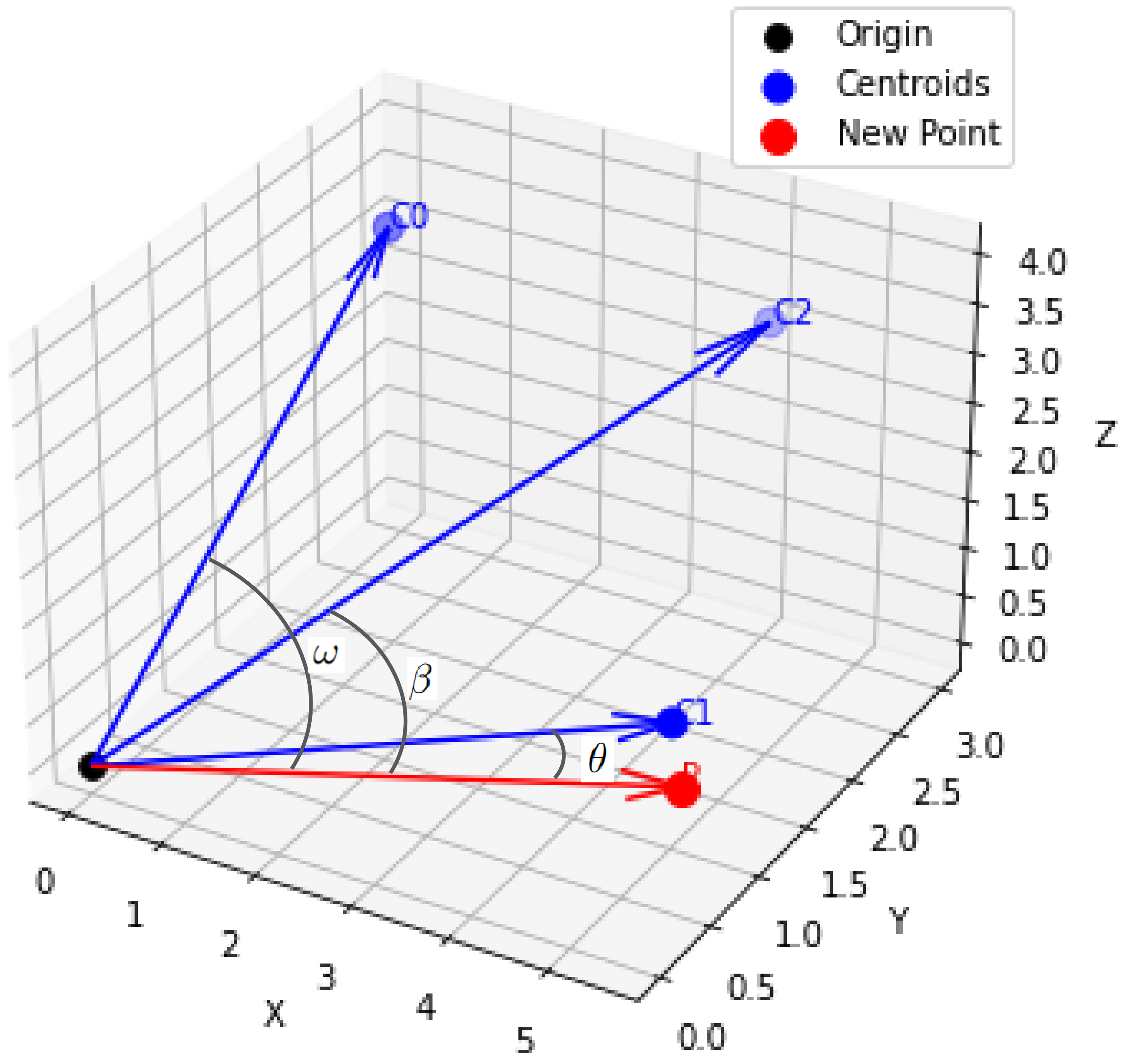
the Hungarian algorithm pair consensus vectors (cluster centroids and mutational signature from a matrix) by maximizing the cosine similarity

$$\mathbf{A} \cdot \mathbf{B} = \|\mathbf{A}\| \, \|\mathbf{B}\| \cos\theta$$

$$\text{cosine similarity} = S_C(A,B) := \cos(\theta) = \frac{\mathbf{A} \cdot \mathbf{B}}{\|\mathbf{A}\|\|\mathbf{B}\|} = \frac{\sum_{i=1}^{n} A_i B_i}{\sqrt{\sum_{i=1}^{n} A_i^2} \sqrt{\sum_{i=1}^{n} B_i^2}},$$

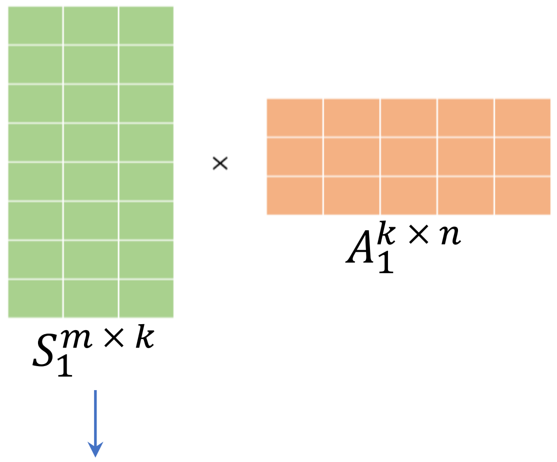


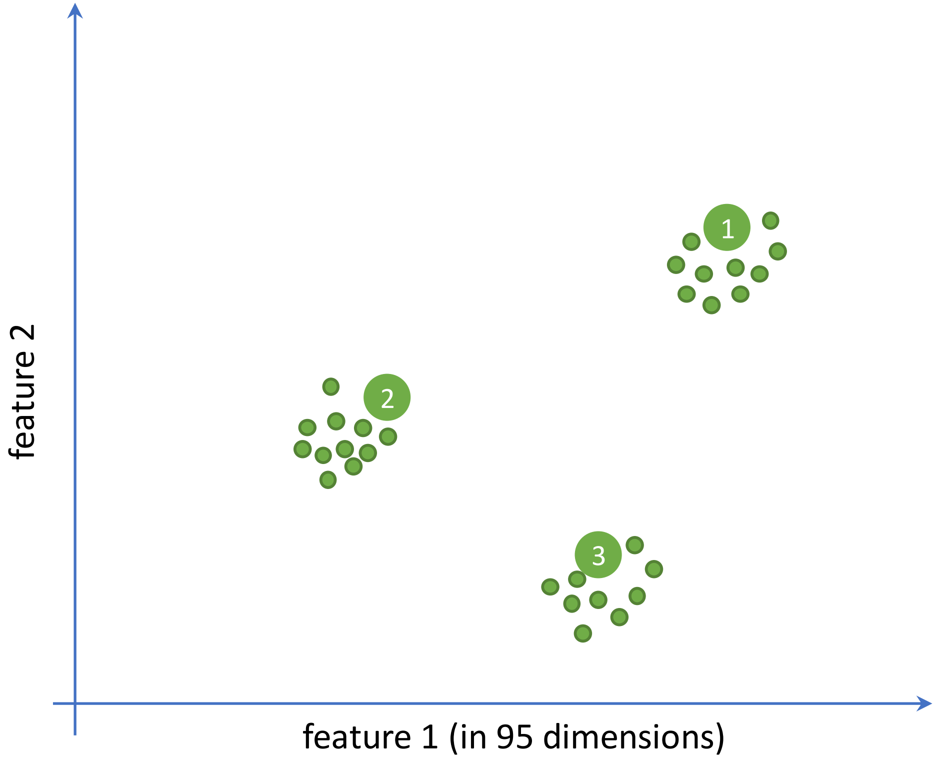



Origin
Centroids
New Point
C0
C2
C1
ω
β
θ
X
Y
Z
0
1
2
3
4
5
0.0
0.5
1.0
1.5
2.0
2.5
3.0
0.0
0.5
1.0
1.5
2.0
2.5
3.0
3.5
4.0

Partition clustering

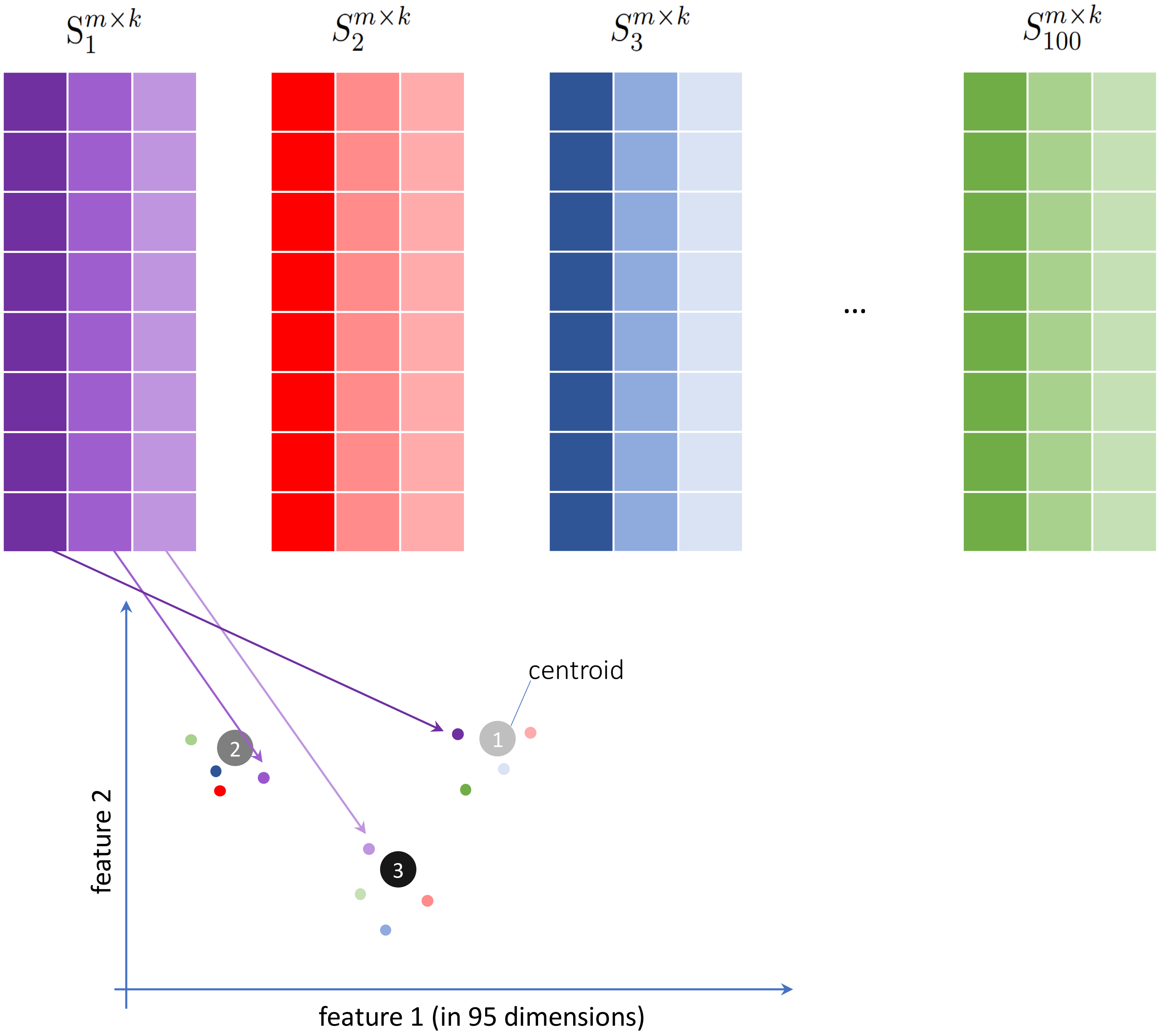
k = 3 (number of operative signatures)





<u>Clustering algorithm</u>

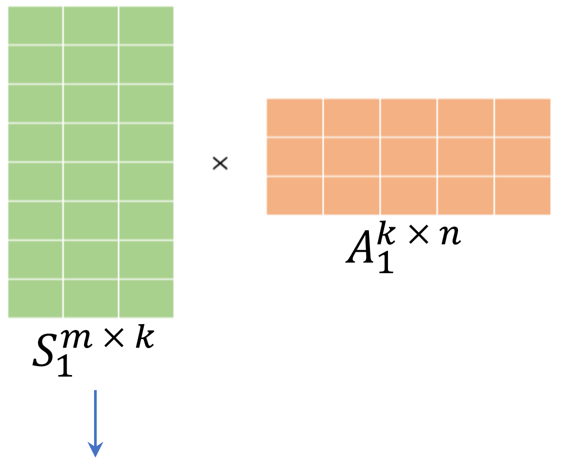
1. k clusters initialized randomly
2. Each column of S_i assigned to a different centroid (in 96 dimensions)
3. **Repeat for all 100 S_i matrices**
4. After assigning all columns to a cluster, the centroid of each cluster is recalculated

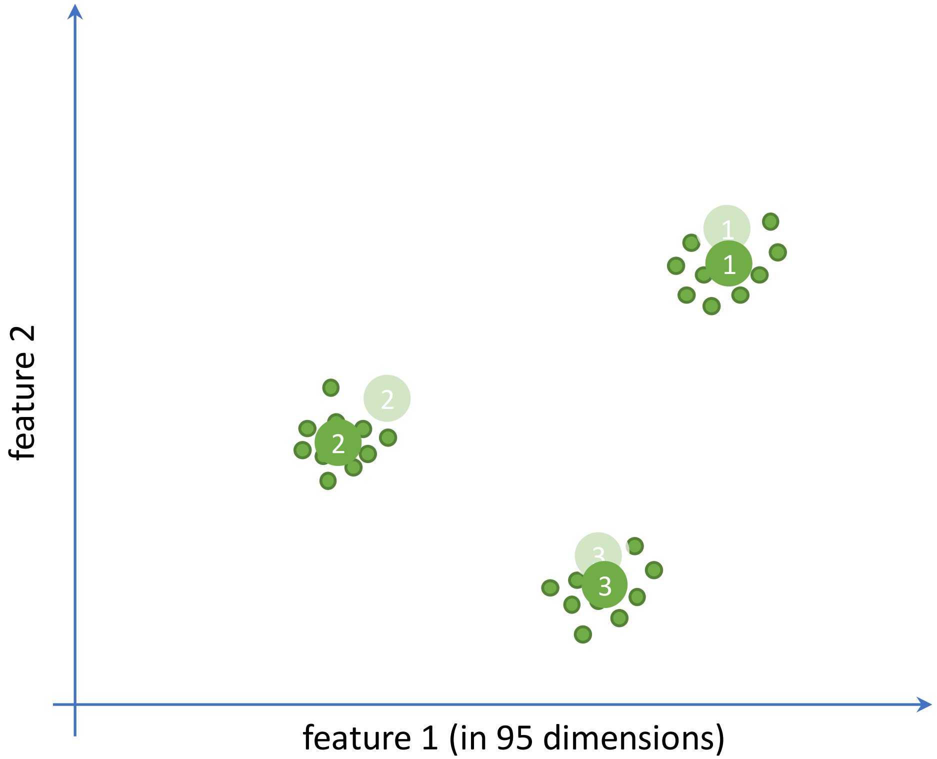


$S_1^{m\times k}$
$S_2^{m\times k}$
$S_3^{m\times k}$
...
$S_{100}^{m\times k}$
centroid
1
2
3
feature 2
feature 1 (in 95 dimensions)

Partition clustering

k = 3 (number of operative signatures)





Clustering algorithm

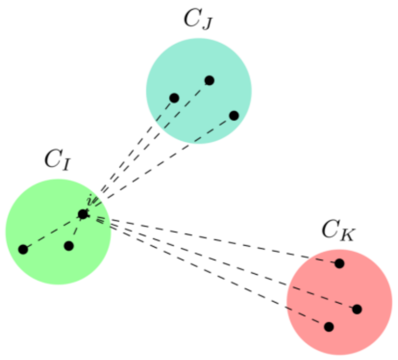
1. k clusters initialized randomly
2. Each column of Si assigned to a different centroid (in 96 dimensions)
3. Repeat for all 100 Si matrices
4. **After assigning all columns to a cluster, the centroid of each cluster is recalculated until convergence**

Partition clustering

This process continues iteratively until the average silhouette coefficient converges

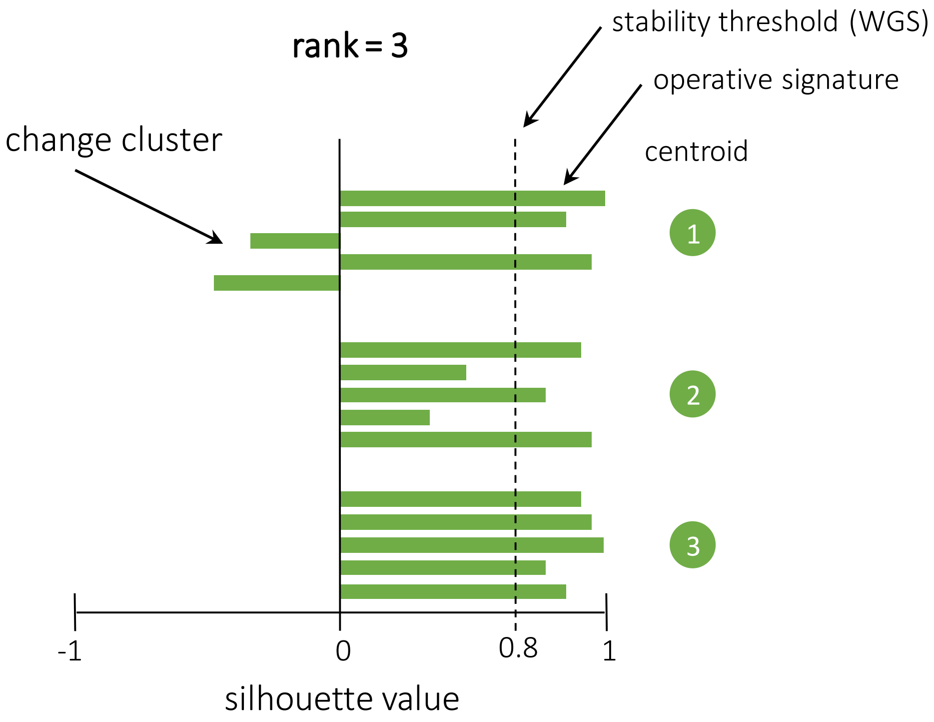
$$a(i) = \frac{1}{|C_I| - 1} \sum_{j \in C_I, i \neq j} d(i,j)$$

$$b(i) = \min_{J \neq I} \frac{1}{|C_J|} \sum_{j \in C_J} d(i,j)$$



<u>silhouette</u>

$$s(i) = \begin{cases} 1 - a(i)/b(i), & \text{if } a(i) < b(i) \\ 0, & \text{if } a(i) = b(i) \\ b(i)/a(i) - 1, & \text{if } a(i) > b(i) \end{cases}$$

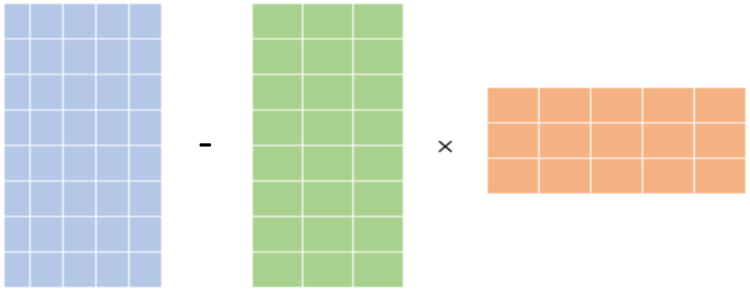


After convergence for a given value of k, the centroids of the clusters are reported as **consensus mutational signatures**

Model selection

Overfitting reduction (i.e. minimize the rank)

reconstructed error (residual) across all samples



rank = 24 R_{24} Wilcoxon rank-sum tests (R_{25}, R_{24})

rank = 23 R_{23} Wilcoxon rank-sum tests (R_{24}, R_{23})

stop when *p-value < 0.05*

⋮ ⋮ ⋮

rank = 1 R_1 Wilcoxon rank-sum tests (R_2, R_1)

The stable solution with the **lowest number of signatures** and a Wilcoxon rank-sum test p-value > 0.05 is selected as the **optimal solution**

Overfitting reduction

For each tested solution (say 10 signatures, 11 signatures, 12 signatures,..), the tool reconstructs each sample's mutational profile using those signatures.

That gives, for sample j:

$$\text{Error}_j^{(k)} = \left\| M_j - \hat{M}_j^{(k)} \right\|$$

where

- M_j : observed mutational spectrum for sample j
- $\widehat{M}_j^{(k)}$: reconstructed spectrum using k signatures
- Error metric : usually *cosine distance* or L_2 norm difference

So for a dataset with N samples you get an array of N errors per solution.

What the Wilcoxon test compares

Let us take the residues obtained with two solutions, for example:

- $E^{(12)}$: reconstruction errors across all samples with 12 signatures → $\left[\overbrace{\mathbf{e}_1^{(12)}, \mathbf{e}_2^{(12)}, \ldots, \mathbf{e}_N^{(12)}}^{\text{samples}}\right]$
- $E^{(11)}$: reconstruction errors across all samples with 11 signatures → $\left[\mathbf{e}_1^{(11)}, \mathbf{e}_2^{(11)}, \ldots, \mathbf{e}_N^{(11)}\right]$

Then apply a Wilcoxon rank-sum test (Mann-Whitney U test):

- H_0: The distribution of errors with 11 signatures is the same as with 12 signatures
- H_A : Errors with 11 signatures are higher (worse fit) than with 12 signatures

Interpretation

- If $p \geq 0.05$: errors are not significantly worse → fewer signatures are acceptable → prefer 11 over 12.
- If $p < 0.05$: errors are significantly worse keep 12.

Then the comparison continues downward (11 vs 10, etc.) until a significant difference is found.

Example with numbers

Imagine 5 samples, errors computed like this:

- 12 signatures: $[0.020, 0.030, 0.010, 0.050, 0.040]$
- 11 signatures: $[0.030, 0.040, 0.020, 0.060, 0.050]$

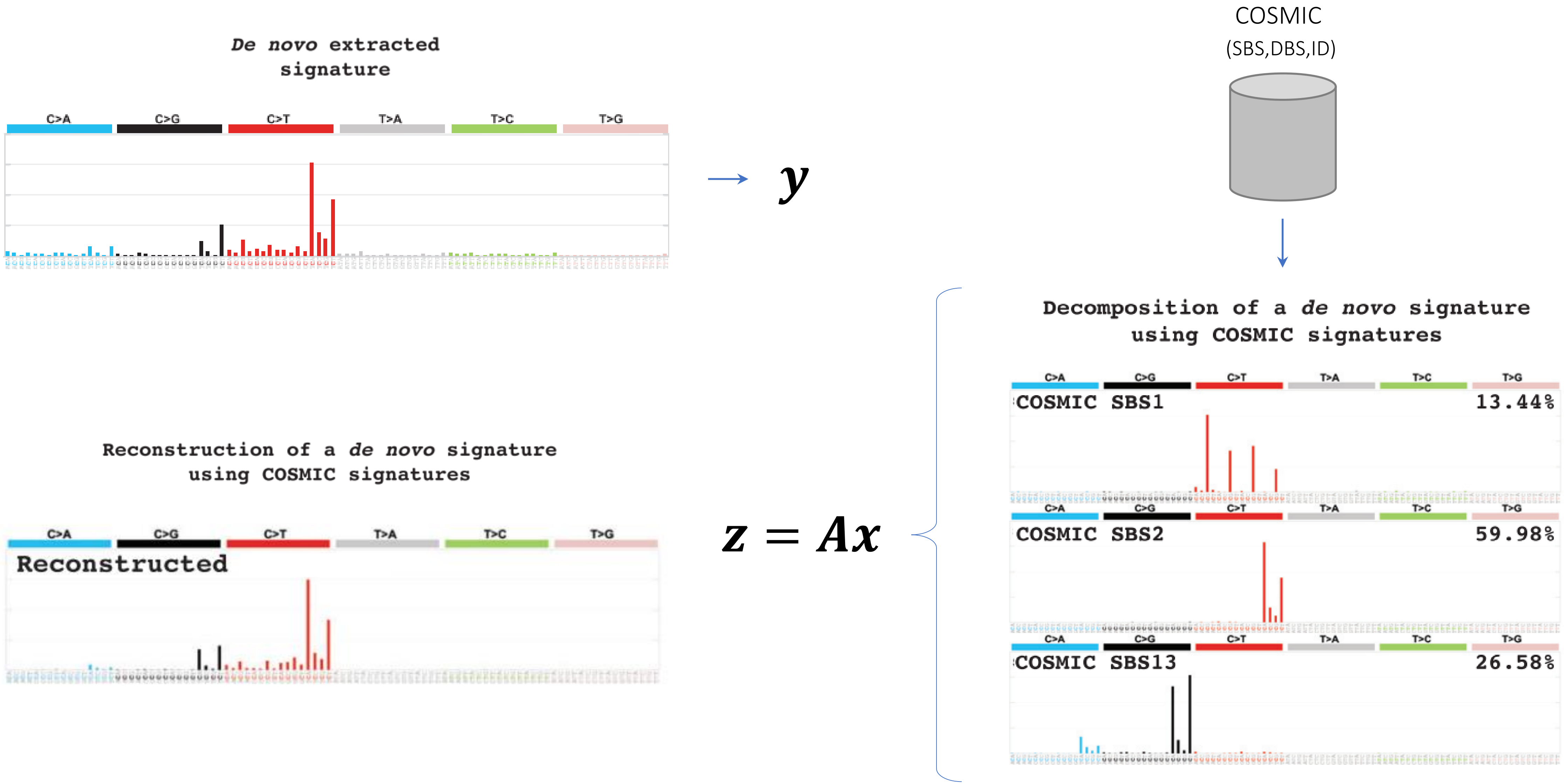
The 11-signature errors are consistently higher. Wilcoxon test will give $p < 0.05$ → reject 11, keep 12.

If instead the arrays were very close:

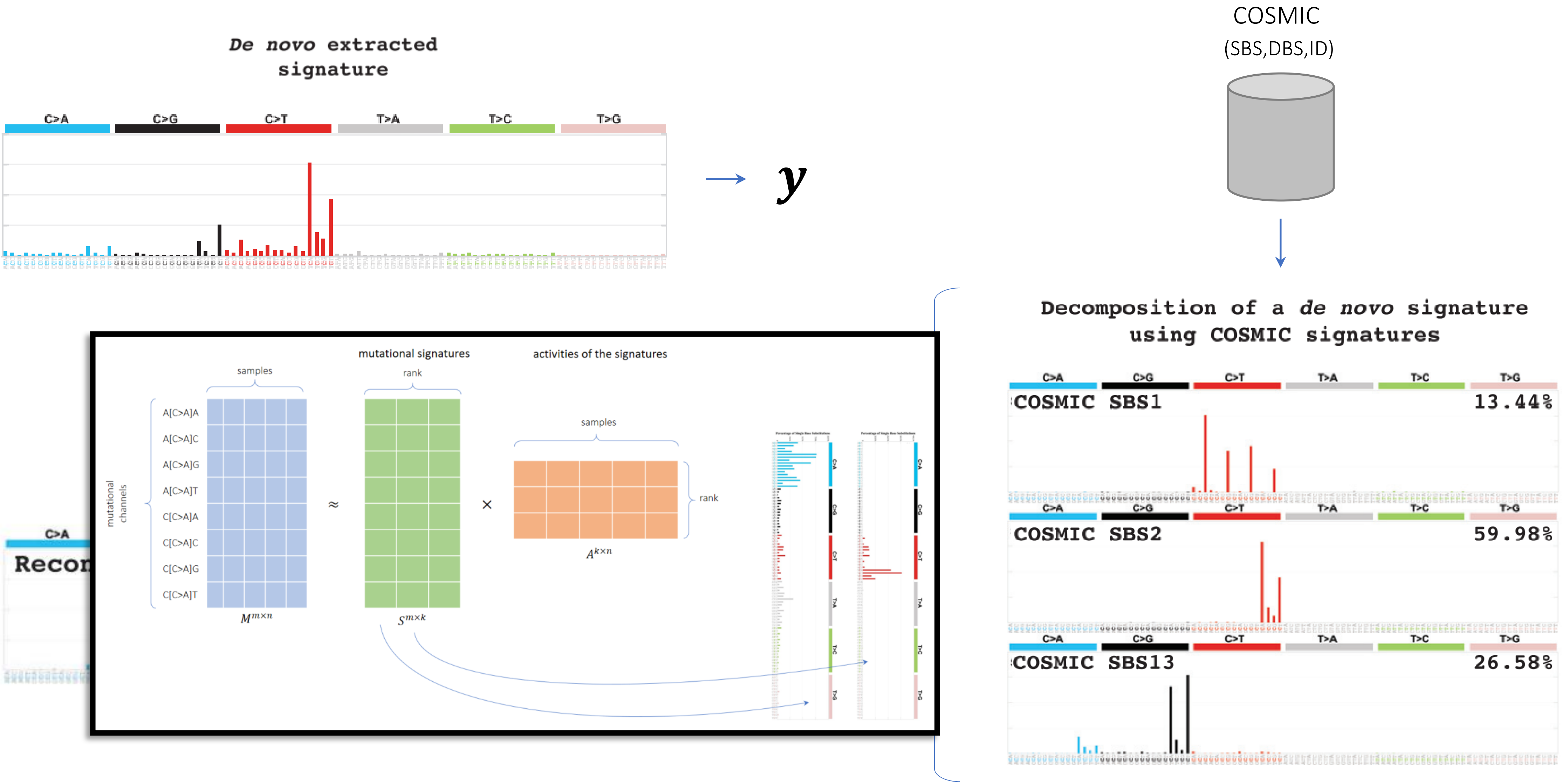
- 12 signatures: $[0.020, 0.030, 0.010, 0.050, 0.040]$
- 11 signatures: $[0.021, 0.031, 0.011, 0.051, 0.041]$

The distributions are nearly identical
$p > 0.05$ keep 11 (simpler).

Decomposing de novo extracted signatures to known COSMIC signatures



Decomposing de novo extracted signatures to known COSMIC signatures



The decomposition functionality leverages the nonnegative least squares (NNLS) algorithm

Minimizes the difference between the extracted signature and the reconstructed signature (using COSMIC signatures)

$$\arg\min_{\mathbf{x}} \underbrace{\|\mathbf{A}\mathbf{x} - \mathbf{y}\|_2^2}_{\text{L2-norm}} \text{ subject to } \mathbf{x} \geq 0$$

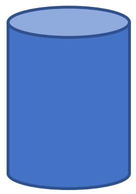
reconstructed signature

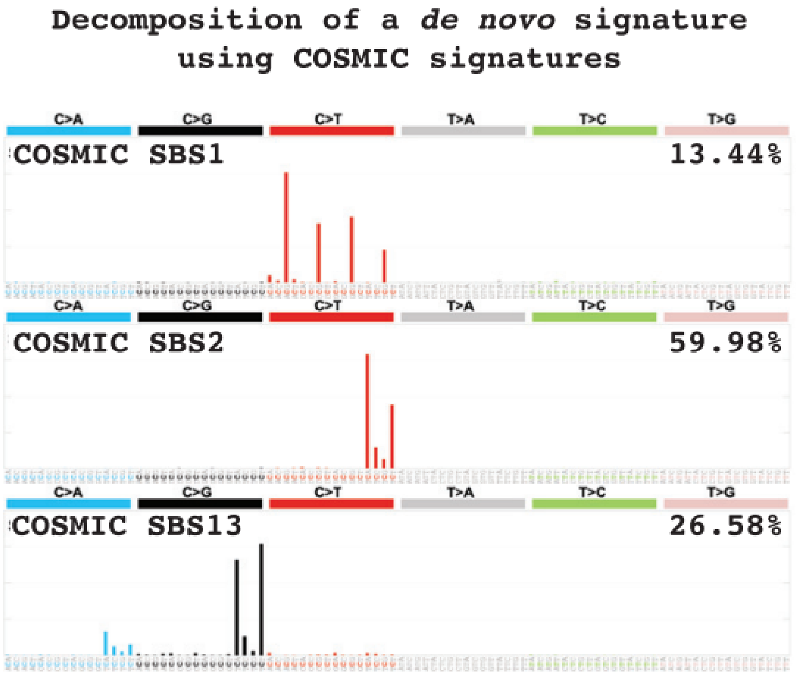
De novo extracted signature

L2-norm

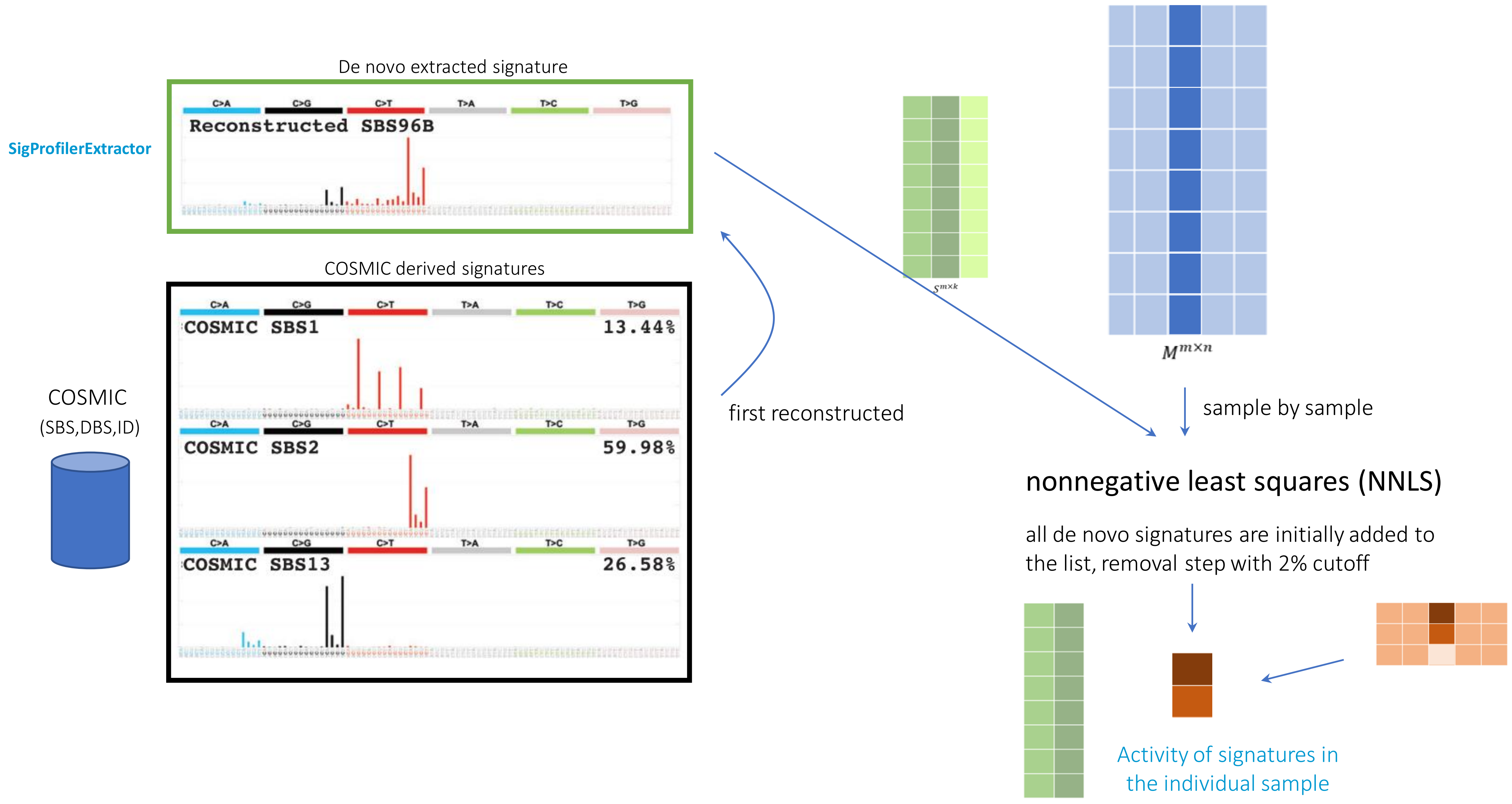
$$\|\mathbf{e}\|_2 = \sqrt{\sum_{k=1}^{n} |e_k|^2}$$

COSMIC
(SBS,DBS,ID)





Evaluating activities of mutational signatures in individual samples



Limitations & caveats

Signatures Are Not Always Unique or Specific

- **Problem**: Mutational signatures **aren't** always **perfect fingerprints**. Sometimes, multiple different biological processes can result in similar-looking mutation patterns. This is called non-uniqueness or **signature crosstalk**.
- **Takeaway**: You can't assume a perfect 1:1 match. For example, signatures related to APOBEC activity (SBS2/13) and some unknown signatures can have overlapping features. Your interpretation must rely on other biological evidence (e.g., APOBEC expression levels in the tumor) to confirm the cause. **The math gives you a match; biology gives you the cause**.

Limitations & caveats

Need Enough Mutations for Reliable Extraction

- **Problem**: Mutational signature analysis (especially the NMF component) is a statistical process that **requires sufficient data**. If a tumor sample has only a very low tumor mutational burden (TMB), the resulting 96-context spectrum is too sparse, making the decomposition unreliable.
- **Takeaway**: The software might give you a result, but it will be noisy and statistically weak. Low-TMB samples often default to ubiquitous signatures like SBS1 (Aging) or SBS5 (Unknown), simply because the unique, high-confidence signature is mathematically hidden by the lack of data. **More data equals higher statistical power**.

Limitations & caveats

Clinical Translation Still Evolving (Not all Signatures are Actionable)

- **Problem**: While the discovery of signatures has been revolutionary, the direct application in the clinic is still limited. **Only a handful of signatures are considered truly actionable** (i.e., predictive of drug response).
- **Takeaway**: You must differentiate between a biologically understood signature and a clinically actionable one.
- **Actionable**: High exposure to SBS3 (HRD) suggests sensitivity to PARP inhibitors. Understood but **Not Always Actionable**: SBS7 (UV light) explains the tumor's origin but doesn't prescribe a specific targeted therapy.
- **Unknown**: Many signatures (like SBS5) have unknown causes and zero clinical use yet.

Limitations & caveats

Noise and Overfitting

- **Problem**: This is a computational risk. When you use NMF to extract signatures de novo (discovery), the algorithm can be influenced by random noise in your small dataset. The result can be overfitted, meaning the extracted signatures explain your specific input data perfectly but lack generalizability to new samples or to the known COSMIC profiles.
- **Takeaway**: **Overfitting leads to finding junk signatures** that are artifacts of random background noise rather than true biological processes. To combat this, researchers rely on established computational metrics, cross-validation, and consensus clustering to ensure the extracted signatures are stable and biologically plausible. Always check your discovered signatures against the COSMIC reference to avoid overfitting to noise.

END